TEMPORAL AWARE STYLE TRANSFER FOR MOTION BASED VIDEO TRANSFORMATION

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CONTENT

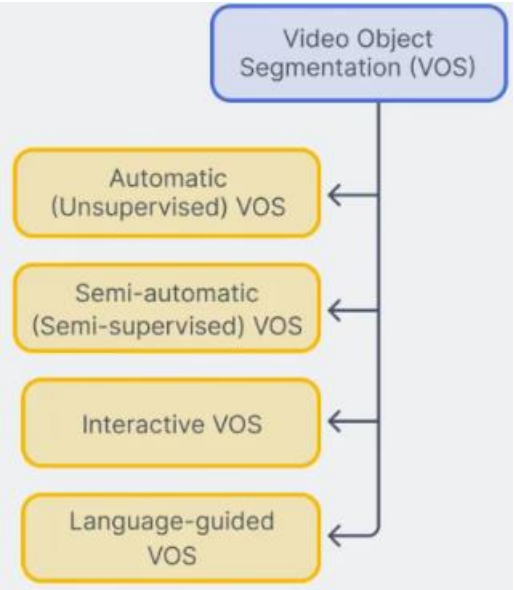
- Introduction
- Research problem and objective
- Literature review
- Research Gap
- Methodology
- Results
- Discussion
- Future work
- Conclusion

INTRODUCTION

- Existing methods treat video frames independently.
  - Introduces flickering, motion artifacts
- Temporal awareness ensures smooth transitions and maintains visual consistency for seamless viewing experience.
- Growing demand in
  - Film making, Gaming, Social media, Virtual reality

Video Object Segmentation

- Generate accurate and temporally consistent pixel masks for objects in a video sequence





G. Bhat, "Learning What to Learn for Video Object Segmentation," Martin Danelljan, Aug. 25, 2020. https://martin-danelljan.github.io/publication/lwtl/

Video Object Segmentation: Challenges

- Strong viewpoint / appearance changes



- Occlusions



Jordi Pont-Tuset, F. Perazzi, and Sergi Caelles, “DAVIS: Densely Annotated VIdeo Segmentation 2017,” *Davischallenge.org*, Apr. 03, 2017. https://davischallenge.org/

Video Object Segmentation: Challenges

- Scale changes



Jordi Pont-Tuset, F. Perazzi, and Sergi Caelles, "DAVIS: Densely Annotated VIdeo Segmentation 2017," *Davischallenge.org*, Apr. 03, 2017. https://davischallenge.org/

- Illumination challenges
- Hard to make assumptions on
  - object motion
  - Object appearance

Style Transfer

- Technique that blends the content of one image with artistic style of another.



RESEARCH PROBLEM

- How can we efficiently apply text guided styles for single or multiple objects in a video, by avoiding temporal inconsistencies, ensuring smooth transition between video frames.

OBJECTIVE

- Develop a deep-learning-based framework for text-guided style transfer that focuses on applying styles exclusively to objects in a video while minimizing temporal inconsistencies between frames

LITERATURE REVIEW

	Research Work	Methodology/ Models	Key findings	Limitations
1	Image style transfer using convolutional neural networks	Use CNN, recombining the content of a photograph with the style of famous artwork to generate high quality new images.	Introduction to the neural algorithm of artistic style transfer. Demonstrate how the CNN can be used to synthesize high quality images by recombining content from photographs and styles from artworks	
2	Artistic style transfer for videos	Optical flow with neural style transfer	Extends style transfer approach for still images to video sequences. Uses optical flow to manage temporal inconsistency.	
3	MOTS: Multi object tracking and segmentation	Combines detection, segmentation and tracking into a unified framework. Includes TrackRCNN baseline method	Creation of MOTS datasets for training and evaluation with instance segmentation and tracking annotations. Introduction of TrackR-CNN, which addresses detection, segmentation, and tracking jointly.	Computationally expensive, struggles with real time applications due to complexity. Struggles to generate masks when occlusions occur.
4	CLIPstyler: Text Driven Style transfer	Style transfer methodology using the CLIP text-image embedding model for semantic texture transfer. Uses patch wise text image matching loss with multiview augmentation with U-Net architecture	Enables style transfer without reference style image.	

LITERATURE REVIEW

5	RVOS: End to end recurrent network for video object segmentation	Propose end to end recurrent network for VOS.	Achieved results on youtube-VOS and DAVIS-2017 benchmarks. Achieves faster inference runtimes.	Depends on the availability of high quality input masks. Performance highly impact in scenes of occlusions and extreme object movements.
6	Multiple style transfer for video for a specific region of interest using deep learning	CNN with the pretrained VGG-16 model	Application of multiple styles to a defined specific region of interest within the video.	Application of the styles only to regions of the video frames, not to objects in the video.
7	Text-Driven Stylization of video objects	Combines dual text prompts with CLIP, uses neural atlas method for maintaining temporal inconsistency	Achieves stylization aligned with text prompts while preserving temporal consistency. Allows fine gained control over styles with varying text specificity.	Computationally expensive. Limits real time usability. Performance in scenarios with occlusions, fast motion or extreme lighting conditions is not fully addressed.
8	Text2LIVE: Text driven layered image and video editing	Introduces zero shot text driven editing by generating and editing layers.	Enables high resolution, localized and semantically meaningful edits to natural images and videos.	Relies on the quality of the image. Struggle with complex scenes.
9	Soulstyler: Using Large Language model to guide image style transfer for target object	Combines large language modals to interpret the text and CLIP to match text with image content. A localized text image block matching loss ensures style transfer for target objects.	Achieves accurate style transfer on specific objects	Focused on Image style transfer. Performance is limited in high dynamic scenes. Computationally expensive.

RESEARCH GAP

- Existing work focus on applying style transfer to
  - Whole image / video frame.
  - OR Single object through whole video frames

- Existing all works use **Neural Atals Layer** to manage Temporal inconsistency,
  - Still focus on single object through out the whole video

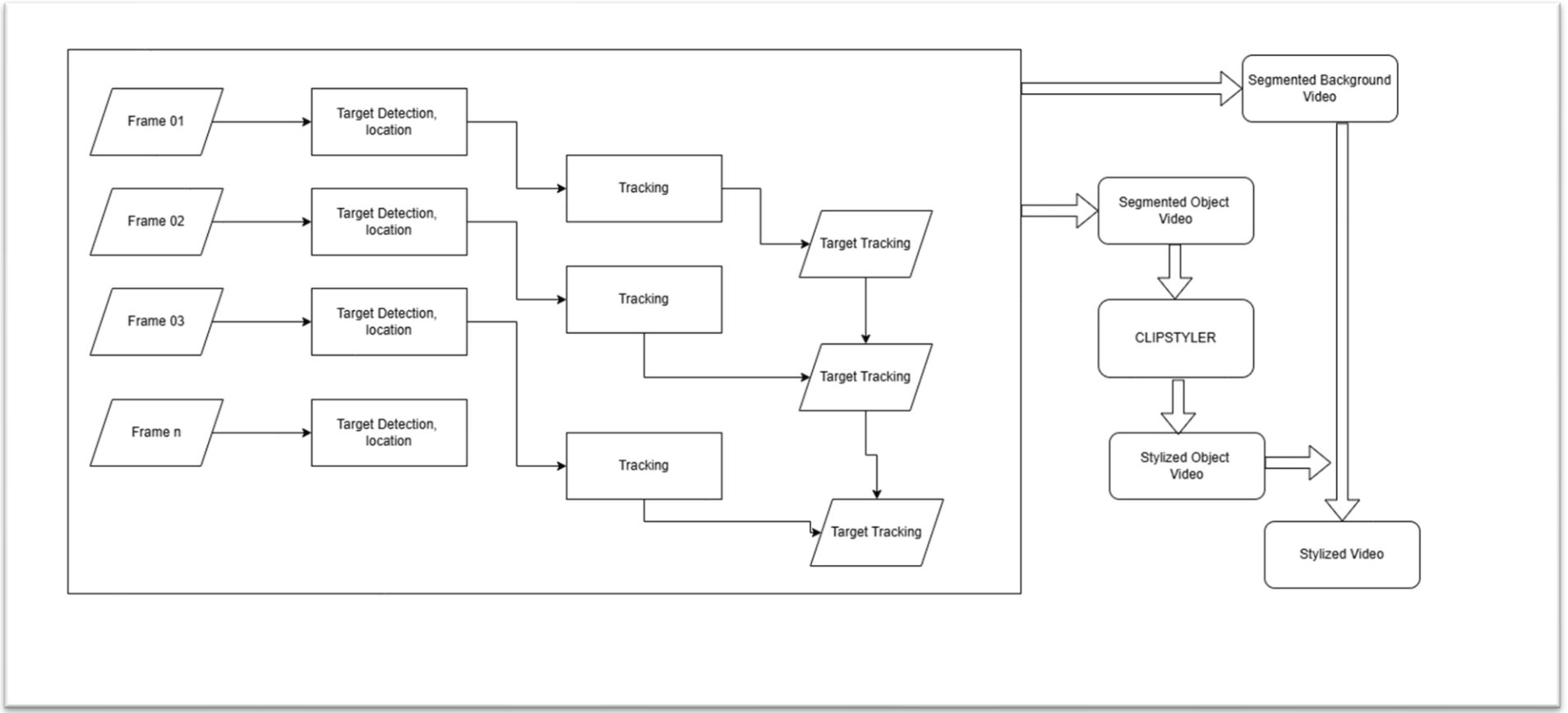
- Need of high computational resources and lacking real time object stylization

AIM OF THE RESEARCH

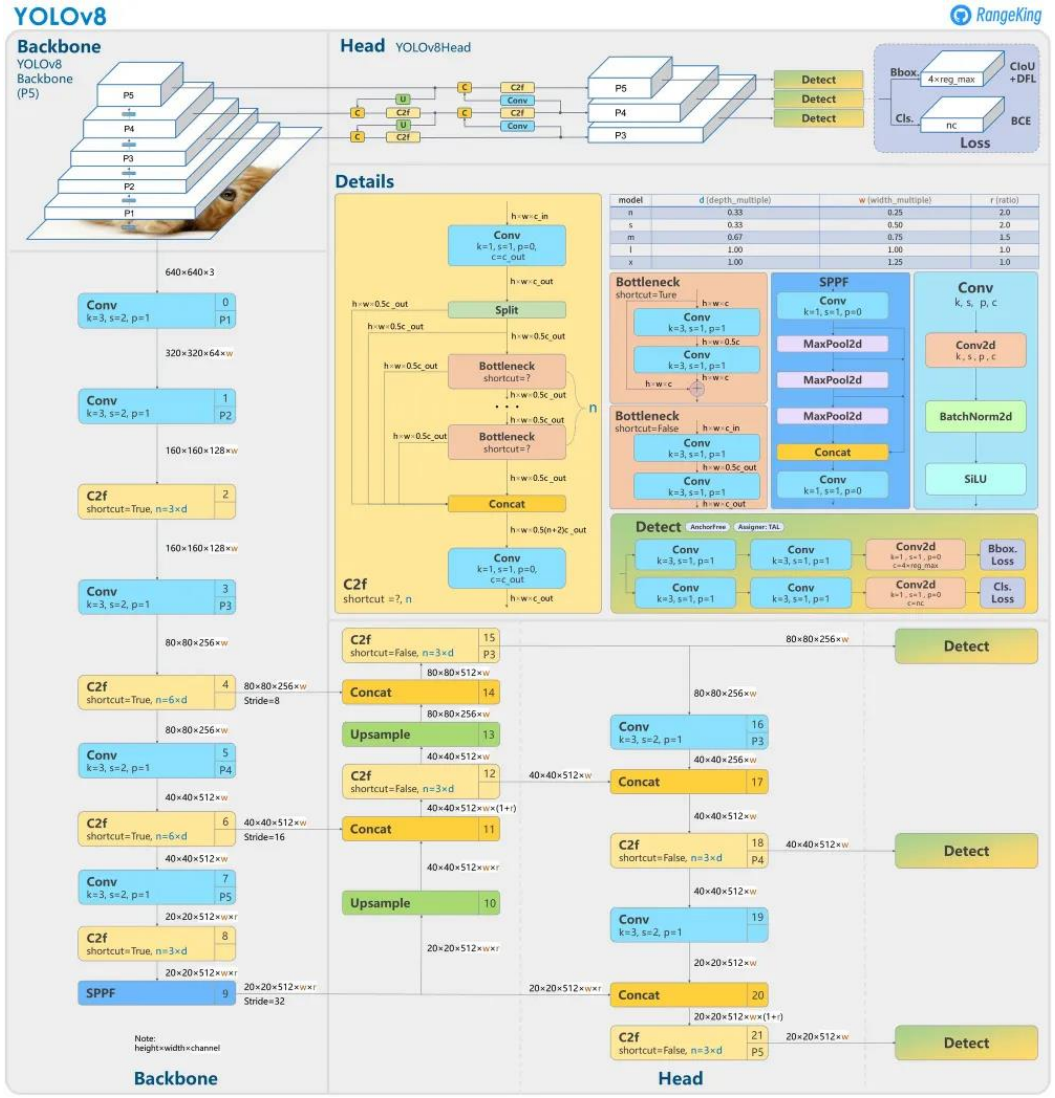
- Develop a style transfer mechanism to transfer styles for single or multiple objects in a video.
- Propose a method to efficiently minimize the temporal inconsistency occurring in between video frames with minimum computational resources.

METHODOLOGY



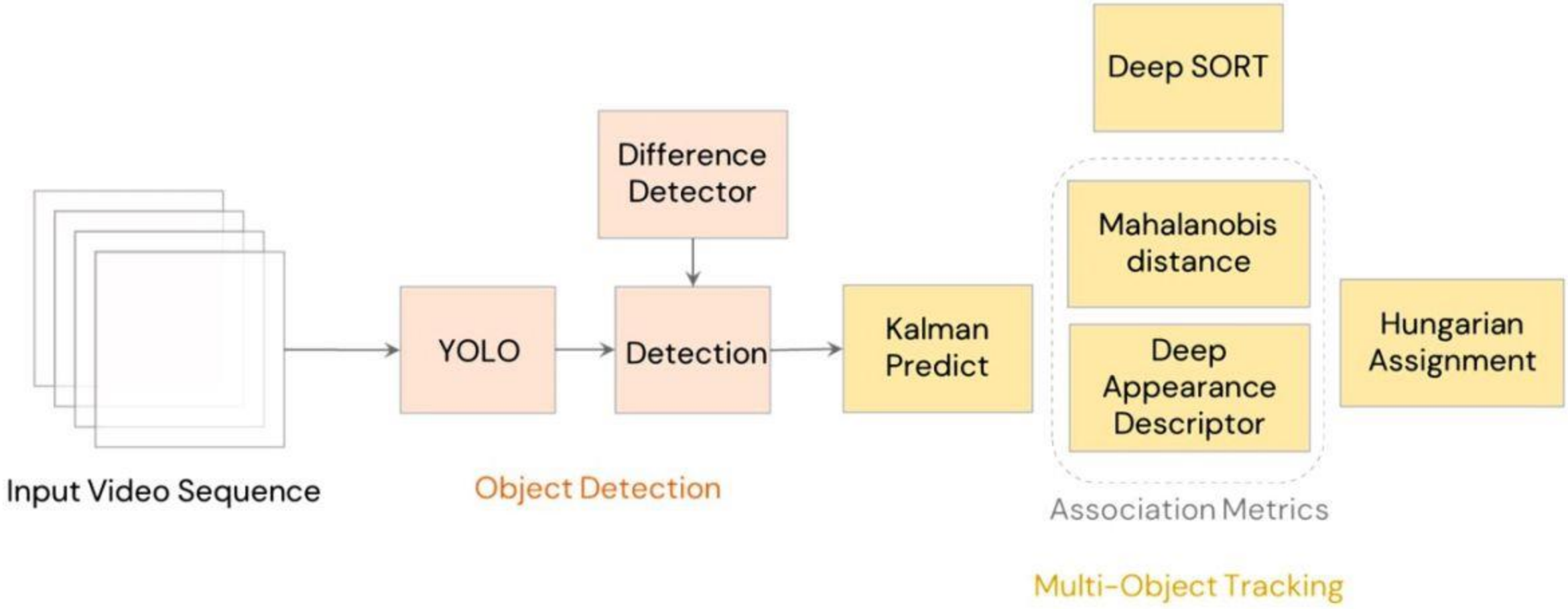


METHODOLOGY : YOLOv8 architecture



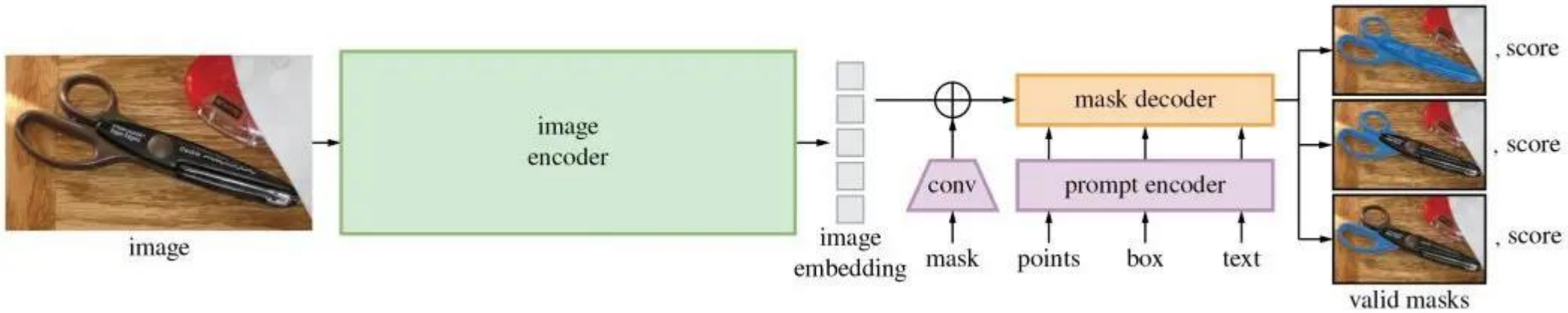
Solawetz, J. (2025, January 10). *What is YOLOv8? A Complete Guide*. Roboflow Blog. https://blog.roboflow.com/what-is-yolov8/

METHODOLOGY : Deep SORT architecture



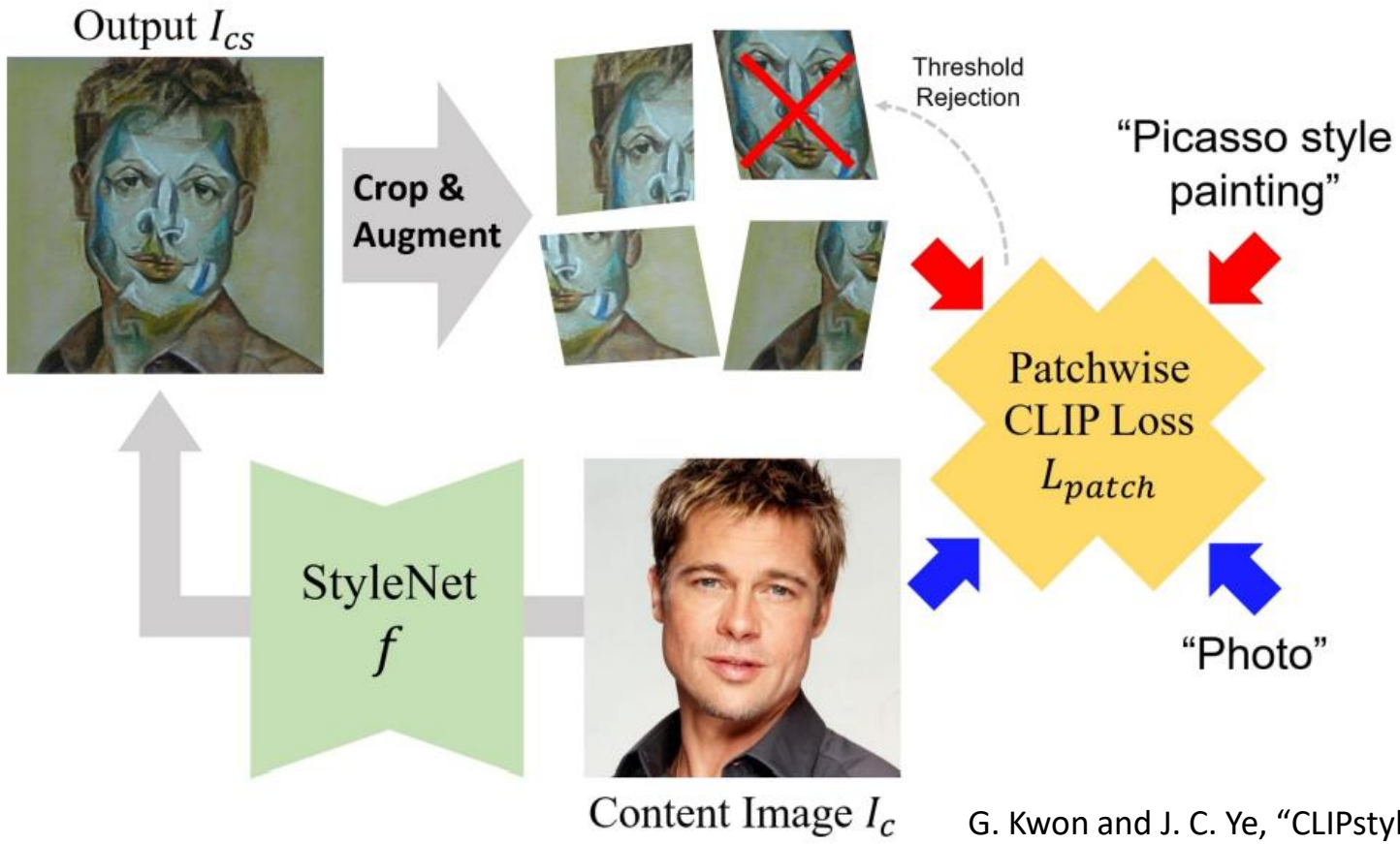
Kouidri, A. (2024, March 7). Deep SORT: Realtime Object Tracking Guide. *Ikomia*. https://www.ikomia.ai/blog/deep-sort-object-tracking-guide

METHODOLOGY : SAM architecture



["Segment Anything | Meta AI," segment-anything.com. [Online]. Available: https://segmentanything.com/

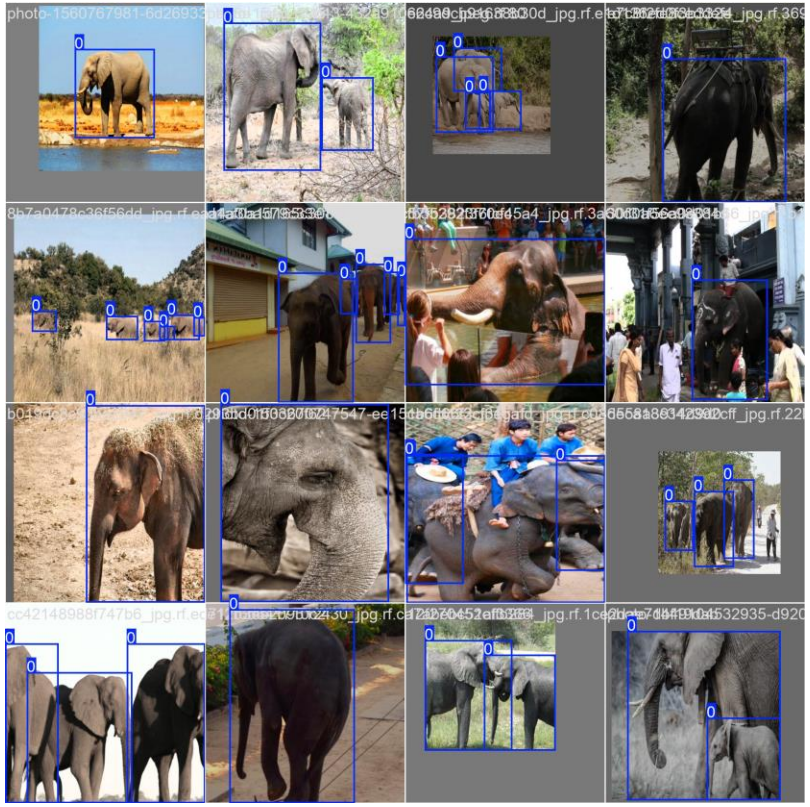
METHODOLOGY : CLIPSTYLER architecture



G. Kwon and J. C. Ye, "CLIPstyler: Image Style Transfer with a Single Text Condition," arXiv.org. [Online]. Available: https://arxiv.org/abs/2112.00374

METHODOLOGY : Dataset

- Dataset
  - Collected through google search engine
  - Total images: 2962 of .jpg format
  - Maximum dimensions of 1600 pixels
  - Range from single to small herds
- Labeling and augmentation
  - Roboflow: Bounding boxes



Required Resources

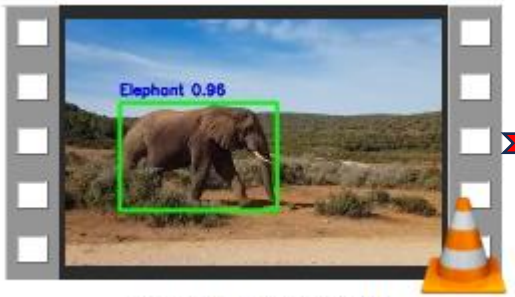
- Hardware requirements
  - GPU : Tesla T4 with 15, 102 MiB memory via Google Colab

- Software requirements
  - Programming language: Python
  - Deep learning models
    - YOLOv8
    - Segment Anything Model
    - CLIPStyler
    - DeepSORT pretrained model

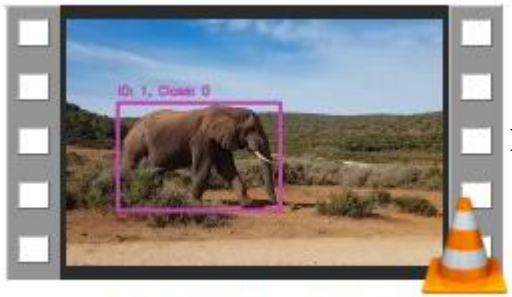
METHODOLOGY : Proposed Approach I



Source Video



OBJECT DETECTION

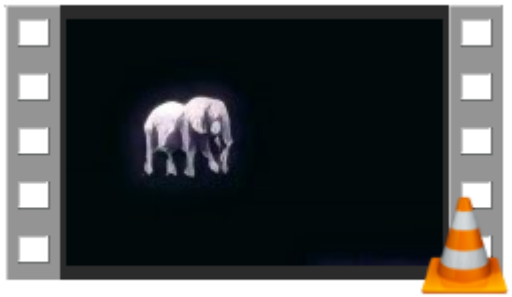


OBJECT TRACKING

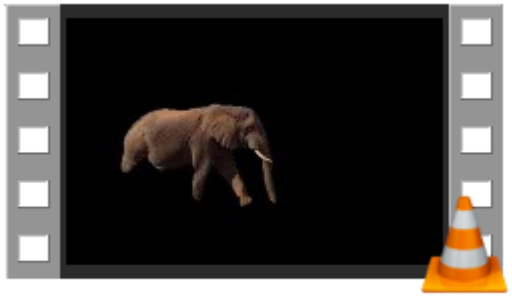


OBJECT SEGMENTATION





STYLIZED FOREGROUNG

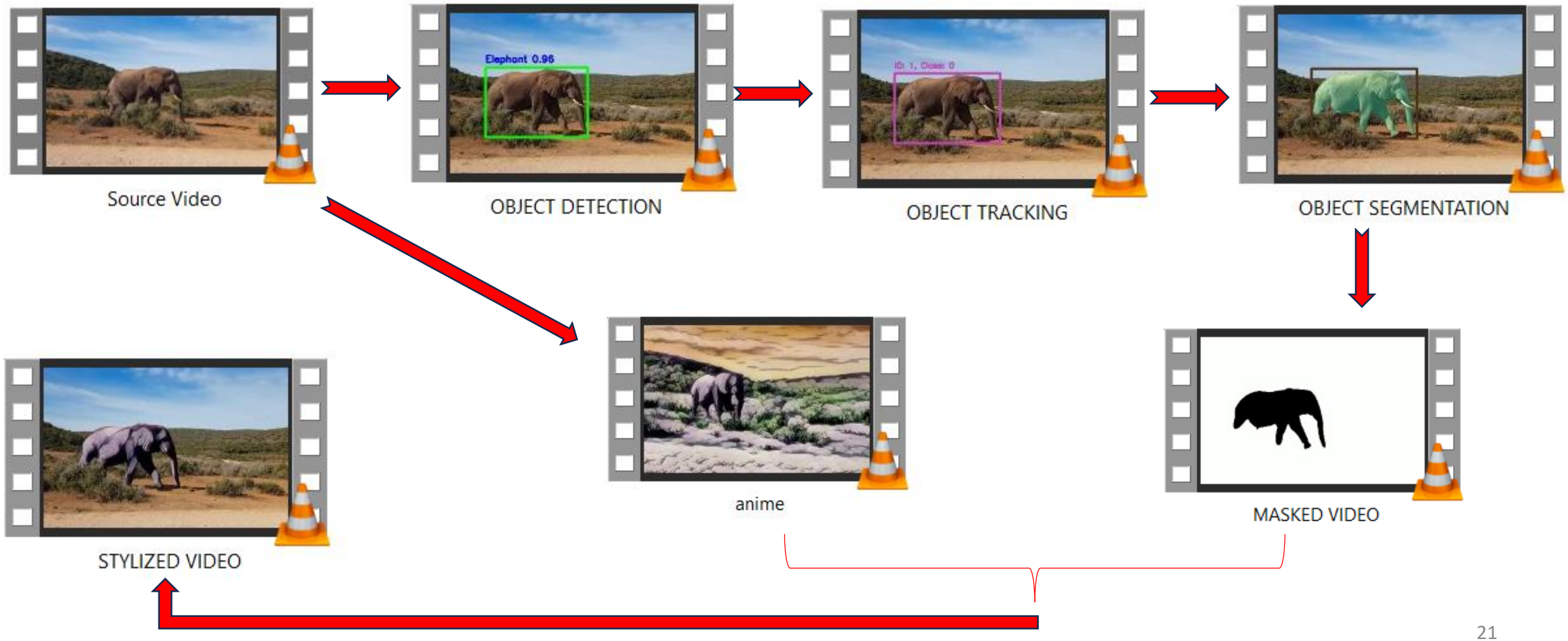


VIDEO FOREGROUND



FINAL STYLIZED VIDEO

METHODOLOGY : Proposed Approach II



EVALUATION

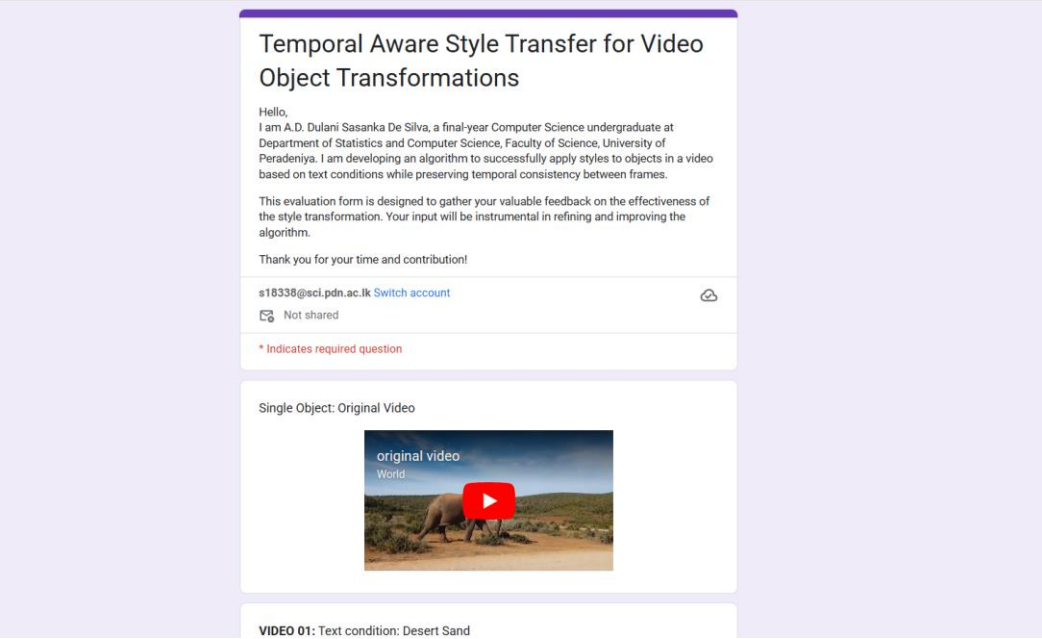
- Evaluation of the models

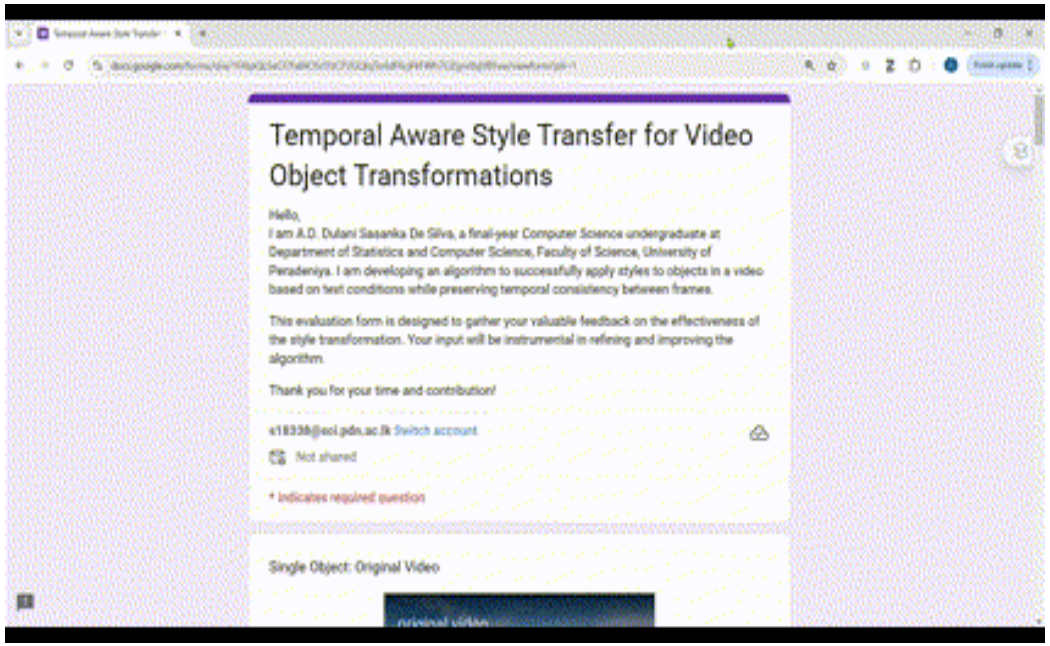
$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \qquad \text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN} \qquad \text{F1-Score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

- Evaluation of the segmentation
  - IOU : Intersection over Union – Measure the overlap between predicted segmentation mask and ground truth
  - Dice Coefficient: Similarity between two sets
  - Precision
  - Recall
  - F1-Score

EVALUATION

- Evaluation of the style transfer for objects in the video



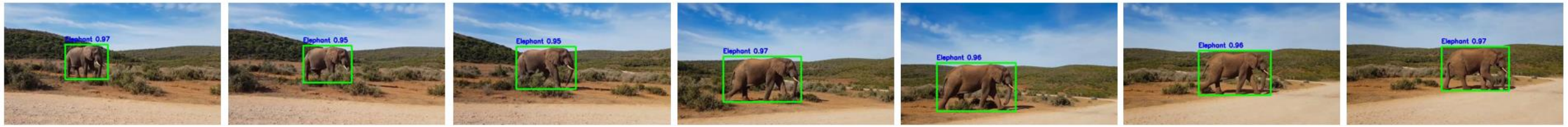


RESULTS OF APPROACH I

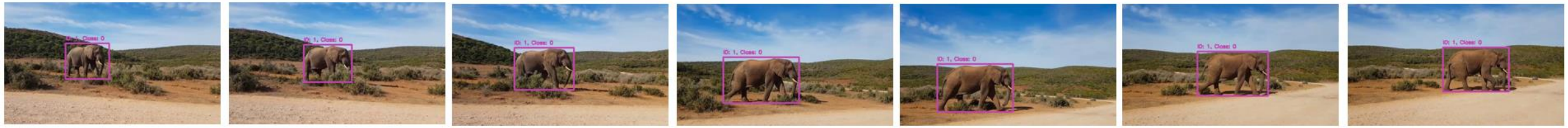
Original video



Object detection



Object tracking



Object segmentation



RESULTS OF APPROACH I

Background only video frames



Foreground only video frames



Stylized foreground video frames: prompt Anime



Final stylized video

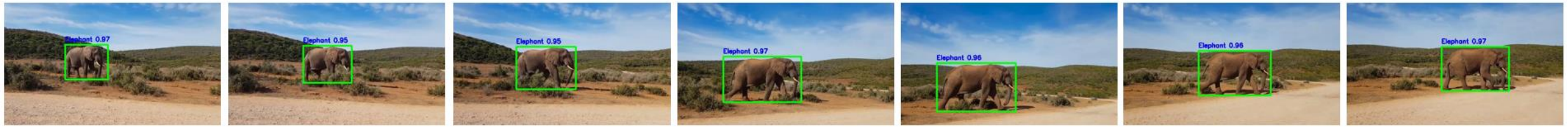


RESULTS OF APPROACH II

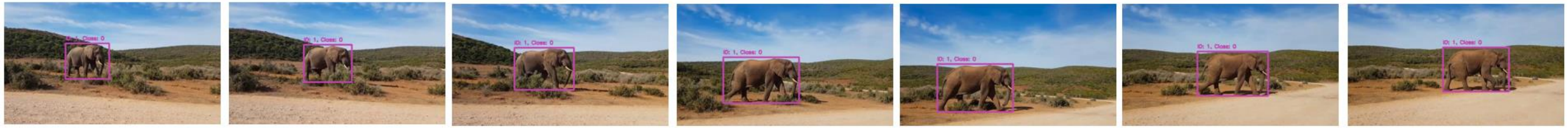
Original video



Object detection



Object tracking



Object segmentation

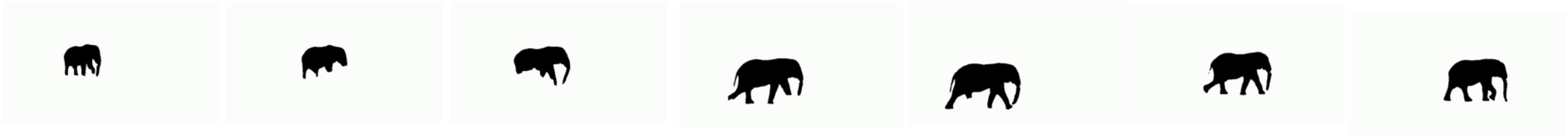


RESULTS OF APPROACH II

Stylized whole video frames: prompt: Anime



Extracted foreground masks



Final Stylized video



APPROACH I vs APPROACH II

Original video



Prompt: Anime

Approach I



Approach II



Prompt: water purple brush

Approach I



Approach II



APPROACH II

Original video



Prompt: Water purple brush





APPROACH I vs APPROACH II

Original video



Prompt: Desert Sand

Approach I



Approach II



Prompt: Black pencil

Approach I

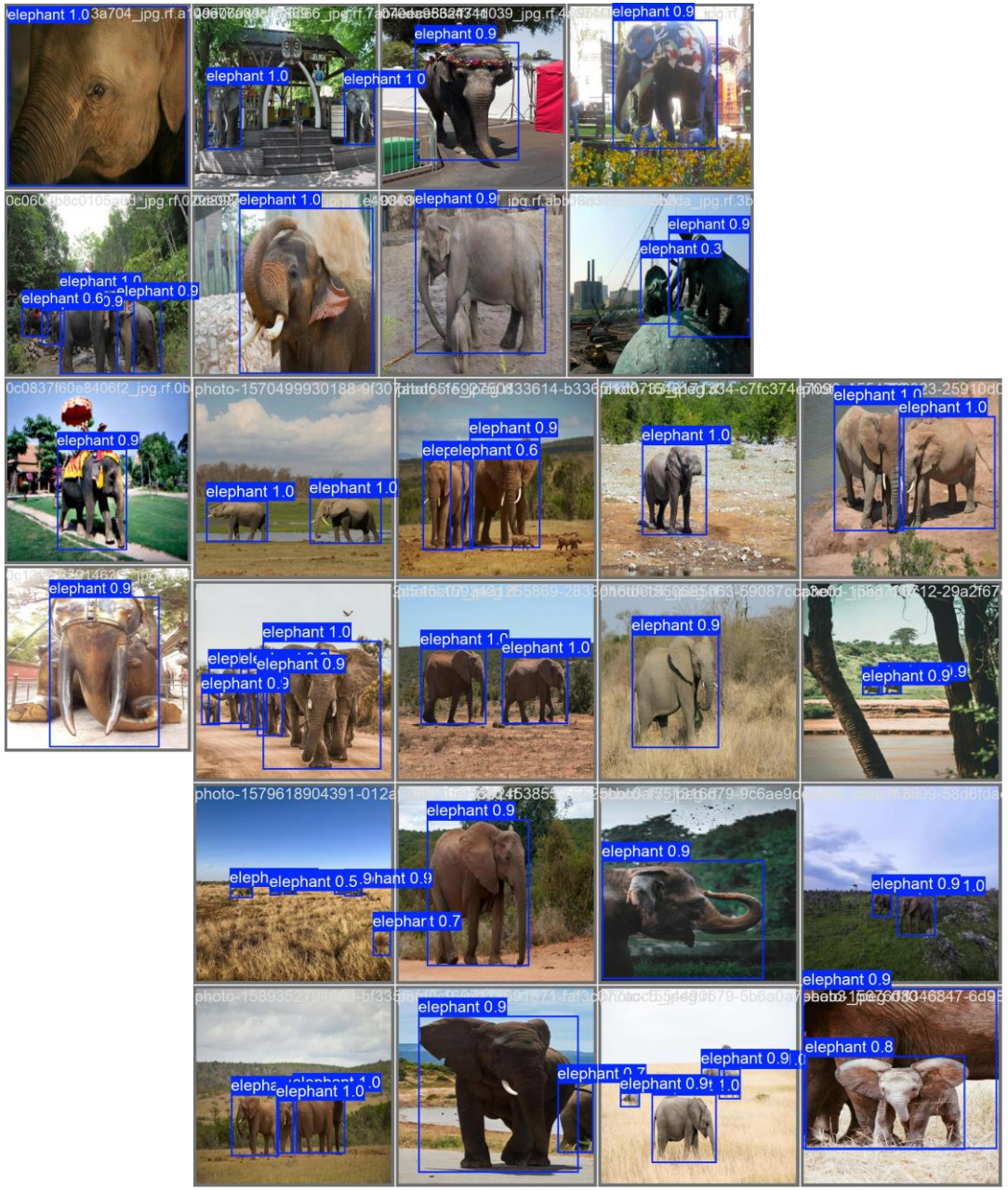


Approach II

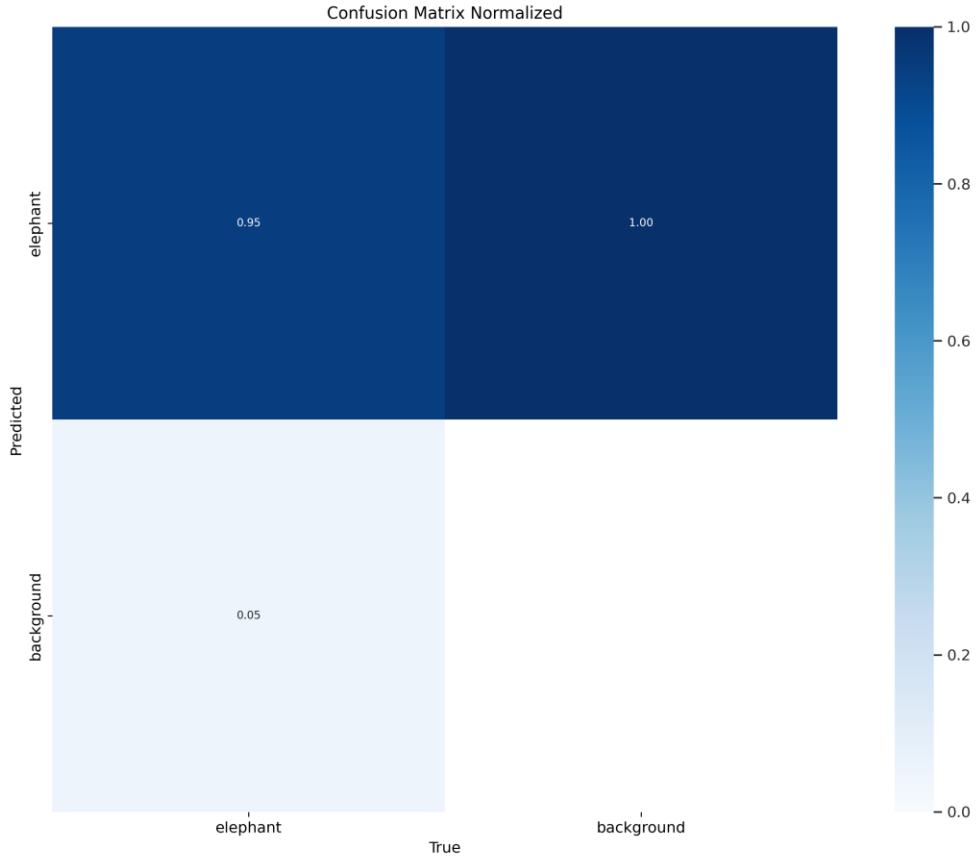


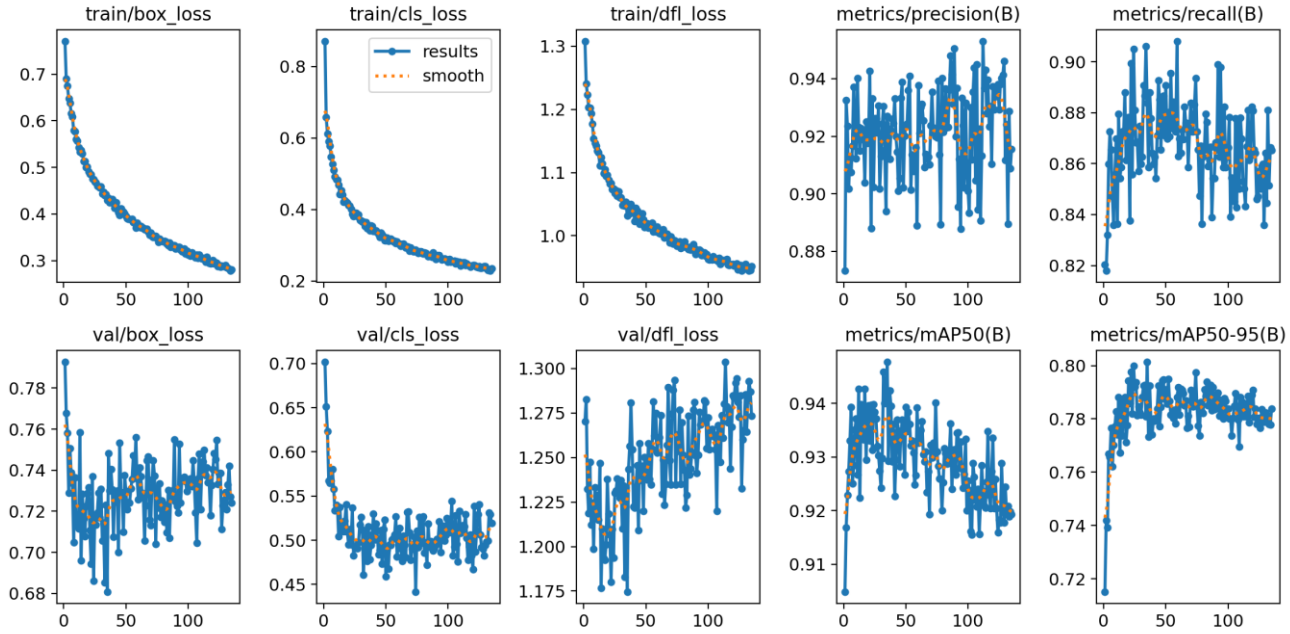
EXPECTED RESULTS

- Precision: 0.90
- Recall: 0.87
- F1-score: 0.89

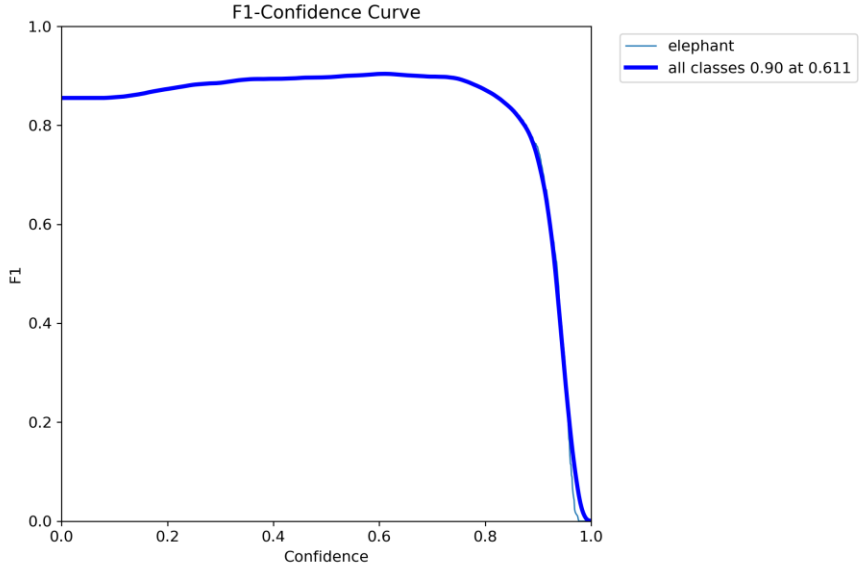


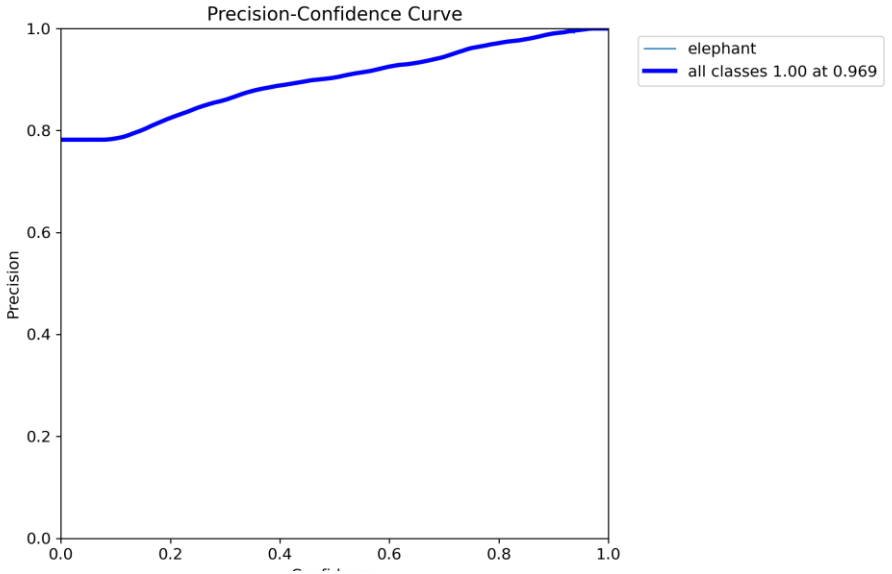
EXPECTED RESULTS

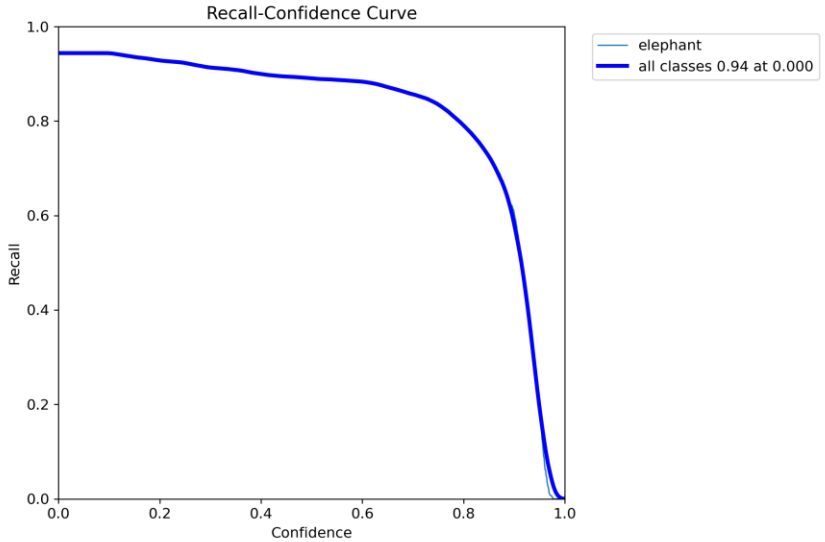


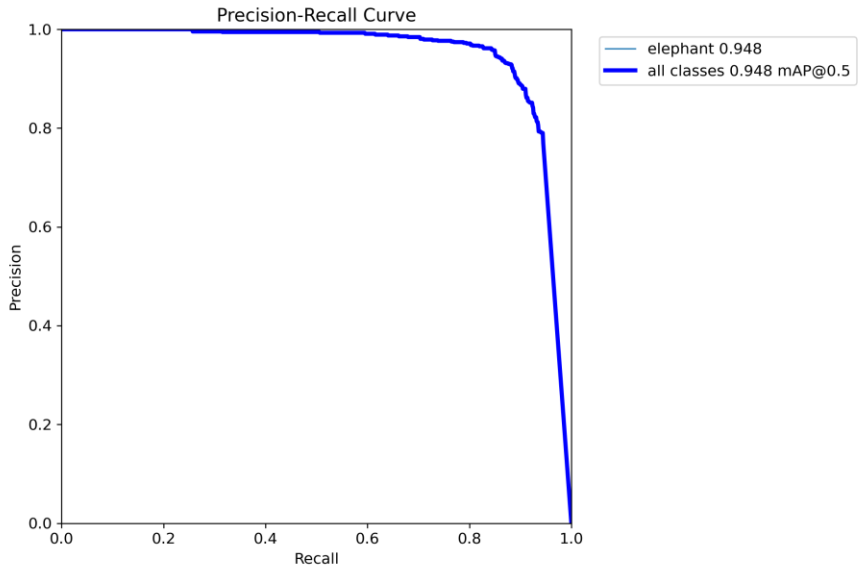


EXPECTED RESULTS

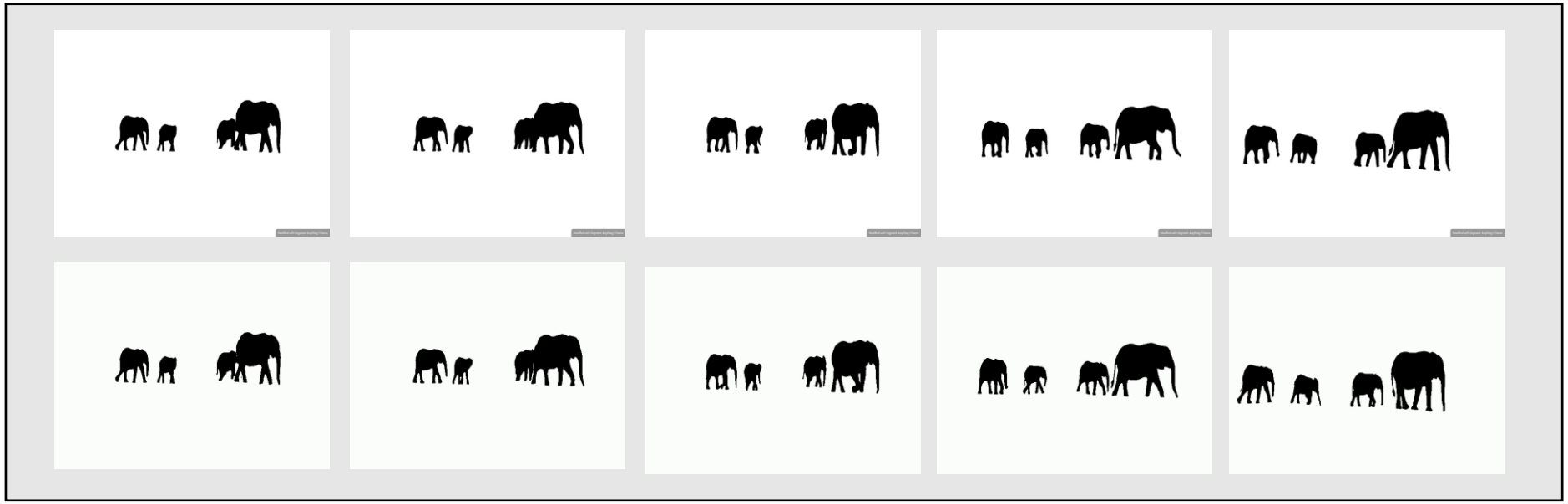








EXPECTED RESULTS



Ground Truth mask obtained through SAM2

Predicted Masks obtained through the methodology

Metric	Mean	Standard Deviation	Median
IOU	0.9681	0.0034	0.9689
DICS	0.9838	0.0018	0.9842
PRECISION	0.9963	0.0011	0.9965
RECALL	0.9715	0.0039	0.9722
F1	0.9838	0.0018	0.9842
BOUNDARY_ERROR	25.9955	10.3855	22.8145
TEMPORAL_CONSISTENCY	0.9957	0.0011	0.9957

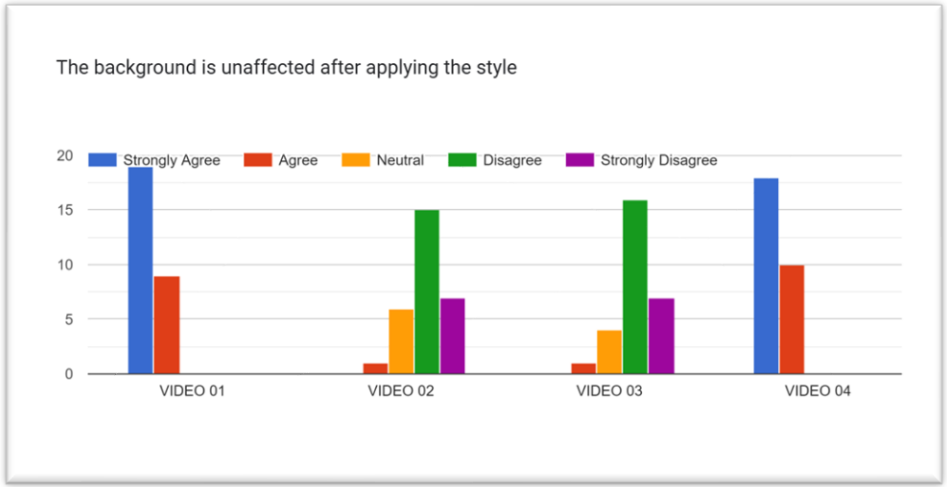
EVALUATION

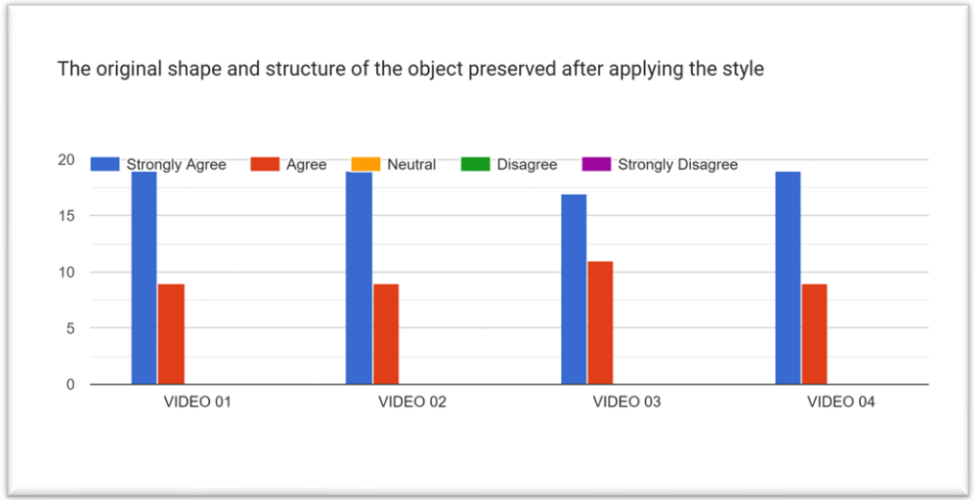

Prompt DesertSand

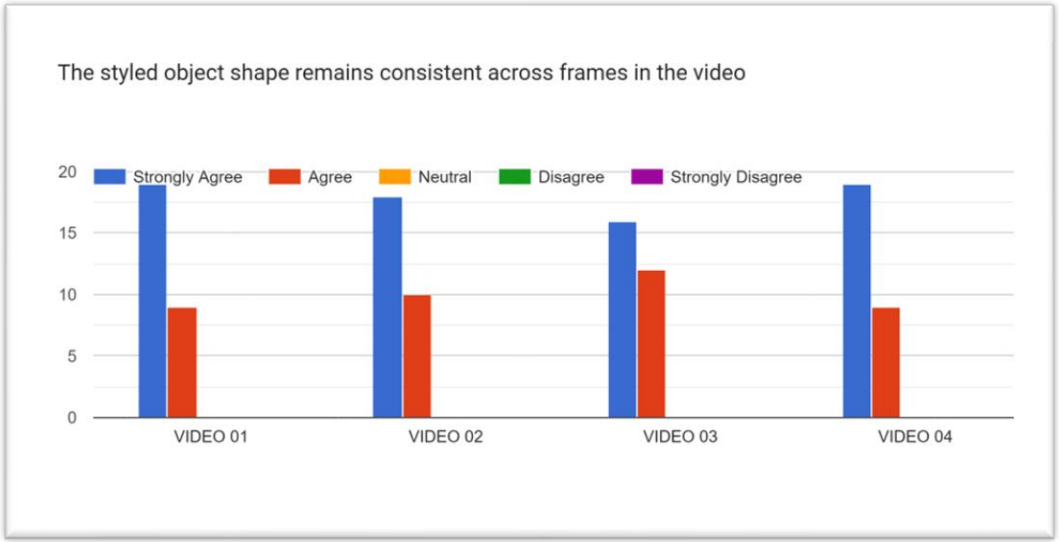

Prompt Anime


Prompt Stonewall


Prompt WaterPurple







EVALUATION



Prompt DesertSand



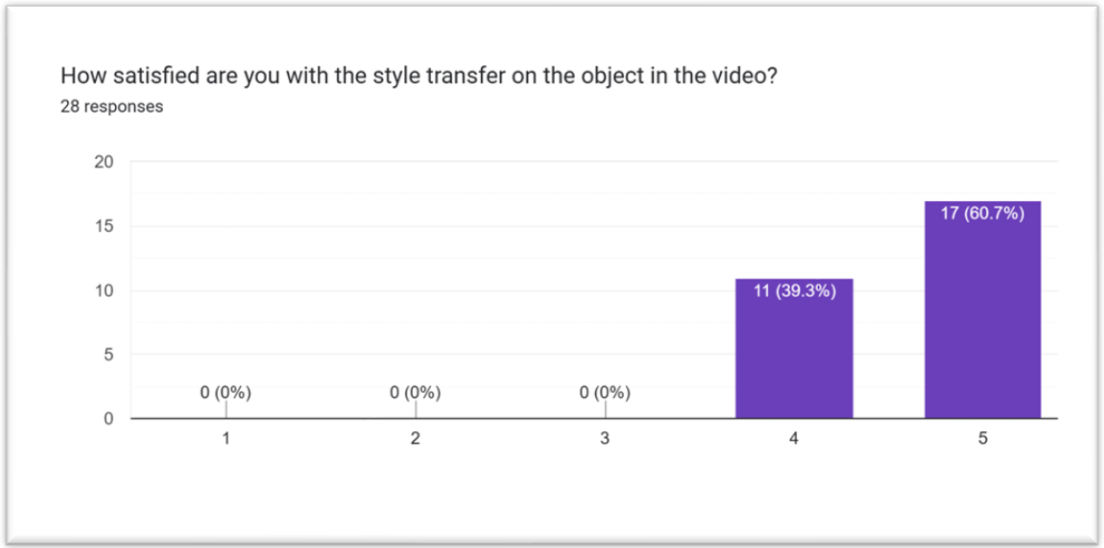
Prompt Anime

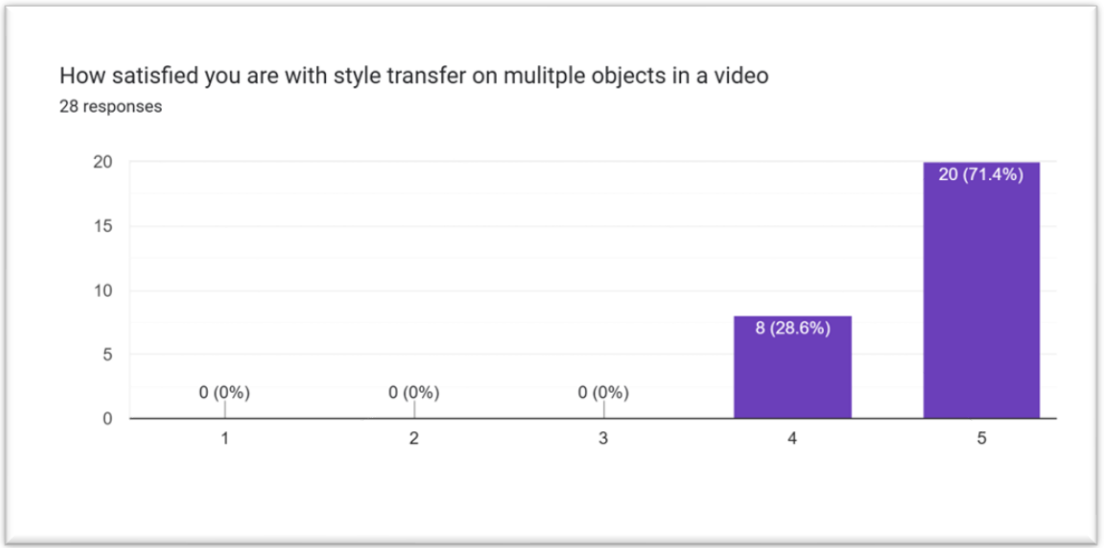


Prompt Stonewall



Prompt WaterPurple





EVALUATION



Prompt DesertSand



Prompt Anime



Prompt Stonewall



Prompt WaterPurple

Question	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
The stylized object matches the describe text condition					
Video 01	67.85%	32.14%			
Video 02	53.57%	46.42%			
Video 03	46.42%	50%	3.57%		
Video 04	67.85%	32.14%			
The styled object shape remains consistent across every frame in the video					
Video 01	67.85%	32.14%			
Video 02	64.28%	35.71%			
Video 03	57.14%	42.85%			
Video 04	67.85%	32.14%			
The original shape and structure of the object preserved after applying the style					
Video 01	67.85%	32.14%			
Video 02	67.85%	32.14%			
Video 03	60.71%	39.28%			
Video 04	67.85%	32.14%			
The background is unaffected by the style applied					
Video 01	67.85%	32.14%			
Video 02		3.57%	21.42%	53.57%	25%
Video 03		3.57%	14.28%	57.14%	25%
Video 04	64.28%	35.71%			

CHALLENGES AND LIMITATIONS



Prompt: Blue Wool

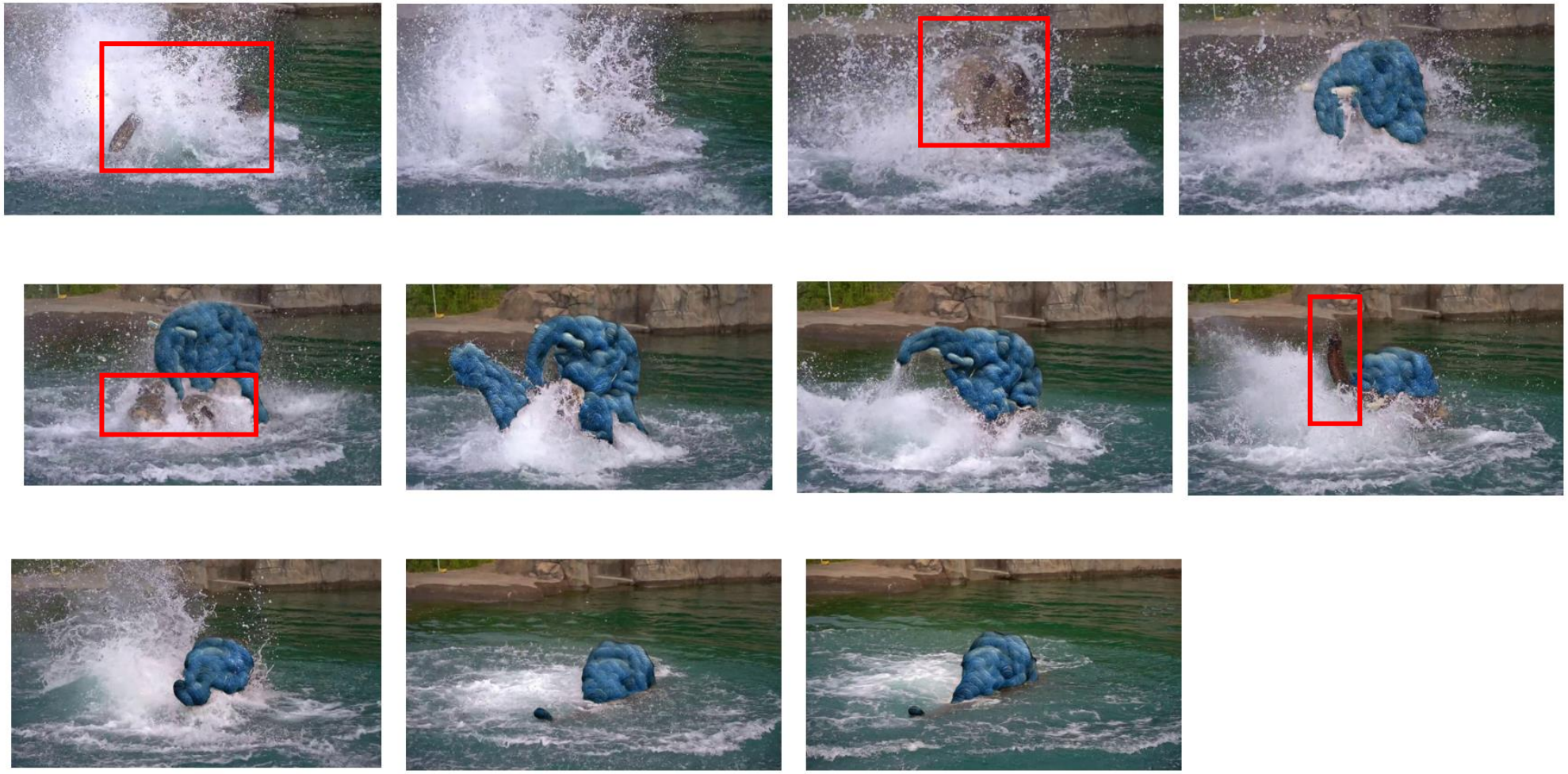
CHALLENGES AND LIMITATIONS

Prompt: Sketch
Black pencil



CHALLENGES AND LIMITATIONS

Prompt: Blue Wool



EXPECTED RESULTS



Ground Truth mask obtained through SAM2

Predicted Masks obtained through the methodology

Metric	Mean	Standard Deviation	Median
IOU	0.0774	0.1766	0.0000
DICS	0.1050	0.2358	0.0000
PRECISION	0.9212	0.1865	1.0000
RECALL	0.3005	0.4199	0.0000
F1	0.2884	0.4084	0.0000
BOUNDARY_ERROR	35.6778	89.2001	0.0000
TEMPORAL_CONSISTENCY	0.9714	0.0747	1.0000

CHALLENGES AND LIMITATIONS

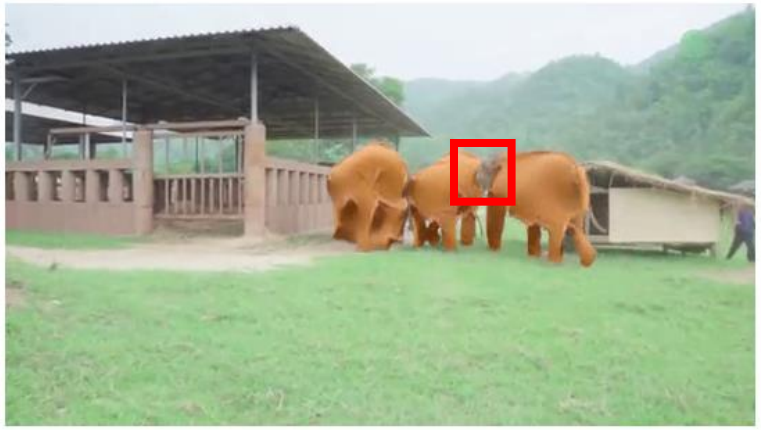
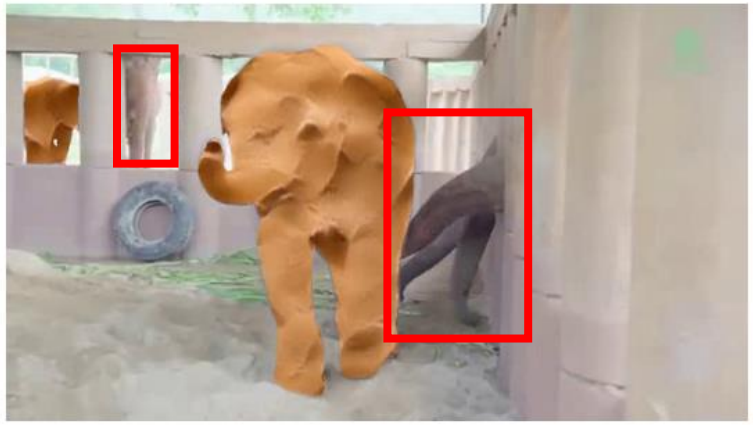
Prompt: Desert Sand










25.00 frames / second

LIMITAITONS OF Proposed Approach

- Dependency on Object detector.
- Impact on video quality
- Struggles to maintain segmentation masks under heavy occlusions.
- May fails in complex environmental conditions

FUTURE WORK

- Replace YOLOv8 with more robust object detection framework
- Integration of SAM2
- Zero shot object detection with advanced integration
  - Combining of SAM2 with and language model (LLM)
- Dynamic multi object styling
  - District styles to multiple objects within same video frame
- Voice command functionality

CONCLUSION

Objective achieved	Transfer styles to single or multiple objects in a video preserving the temporal consistency.
Key Contributions	Proposed an alternative to the neural atlas layer. Implementation of style transfer for single and multiple objects in a video. Manually collected and annotated the dataset of object class Elephant.
Methodology	Focused on single object class YOLOv8, SAM, DeepSORT with CLIPStyler
Outcomes	Achieving of smooth consistent style propagation across video frames
Limitation:	May fail in complex environmental conditions with high occlusions.

REFERENCES

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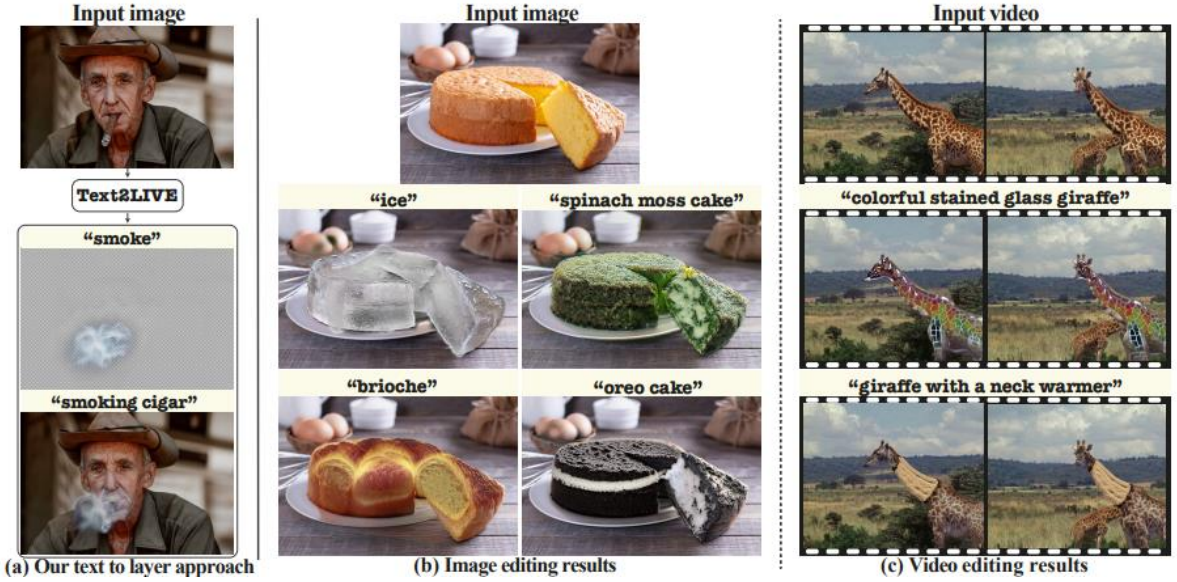
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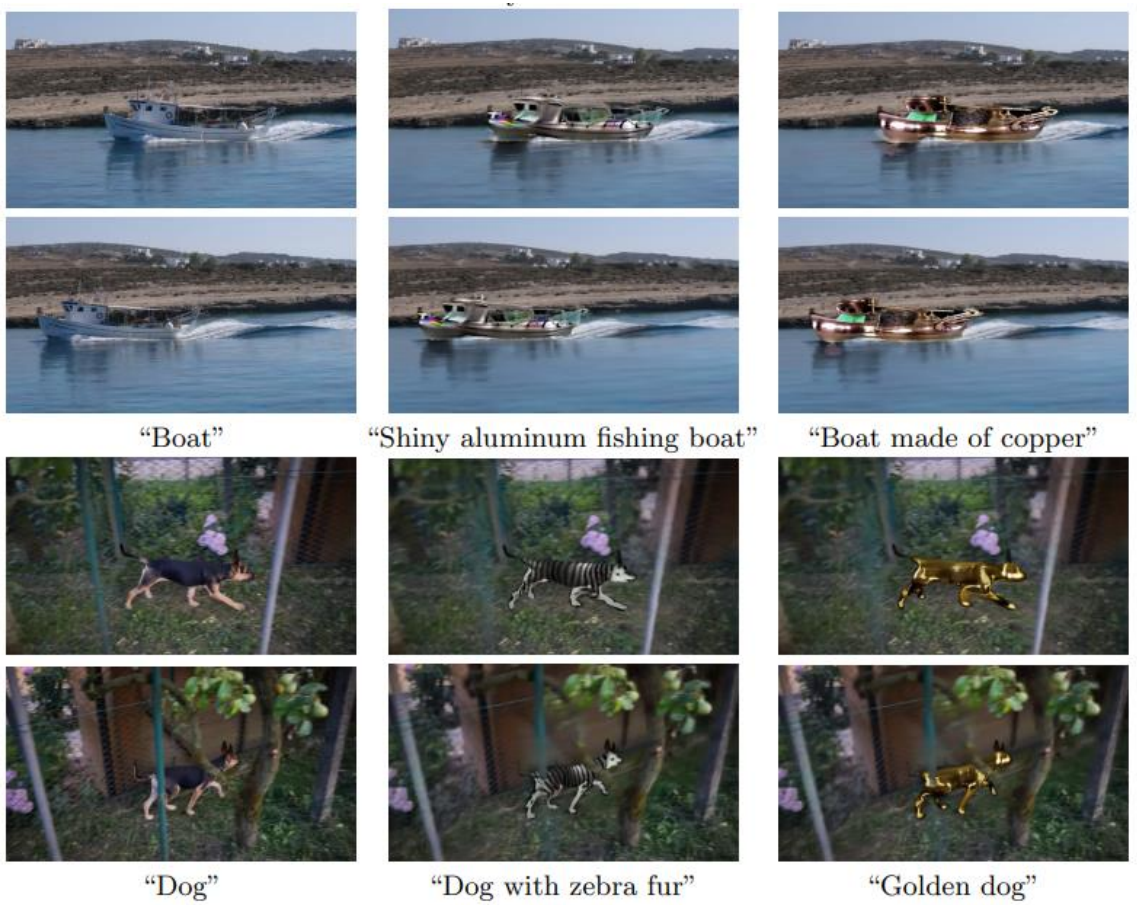
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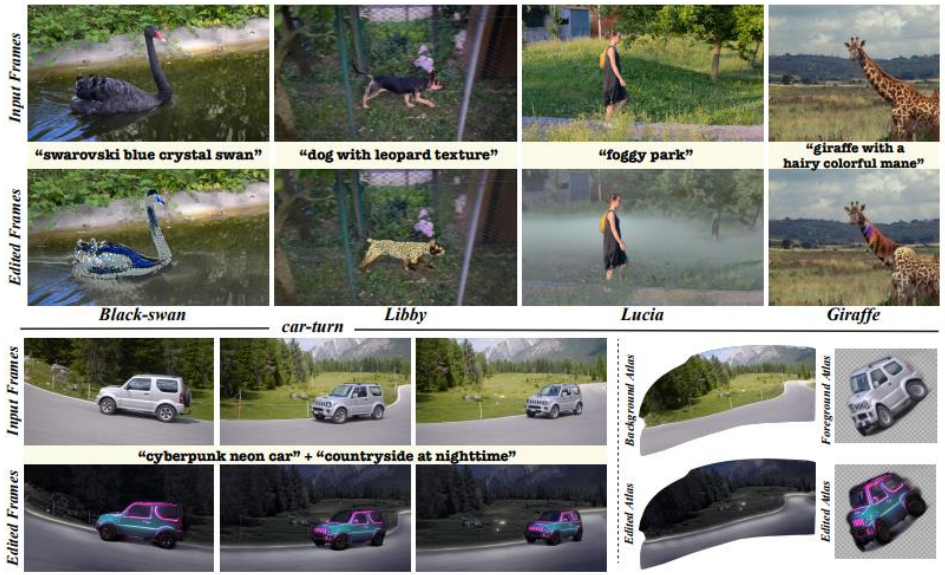
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Thank you !


Input image
Text2LIVE
"smoke"
"smoking cigar"
(a) Our text to layer approach
Input image
"ice"
"spinach moss cake"
"brioche"
"oreo cake"
(b) Image editing results
Input video
"colorful stained glass giraffe"
"giraffe with a neck warmer"
(c) Video editing results


"Boat"
"Shiny aluminum fishing boat"
"Boat made of copper"
"Dog"
"Dog with zebra fur"
"Golden dog"


Input Frames
"swarovski blue crystal swan"
"dog with leopard texture"
"foggy park"
"giraffe with a hairy colorful mane"
Edited Frames
Black-swan
Libby
Lucia
Giraffe
car-turn
Input Frames
"cyberpunk neon car" + "countryside at nighttime"
Edited Frames
Background Atlas
Foreground Atlas
Edited Atlas
Edited Atlas

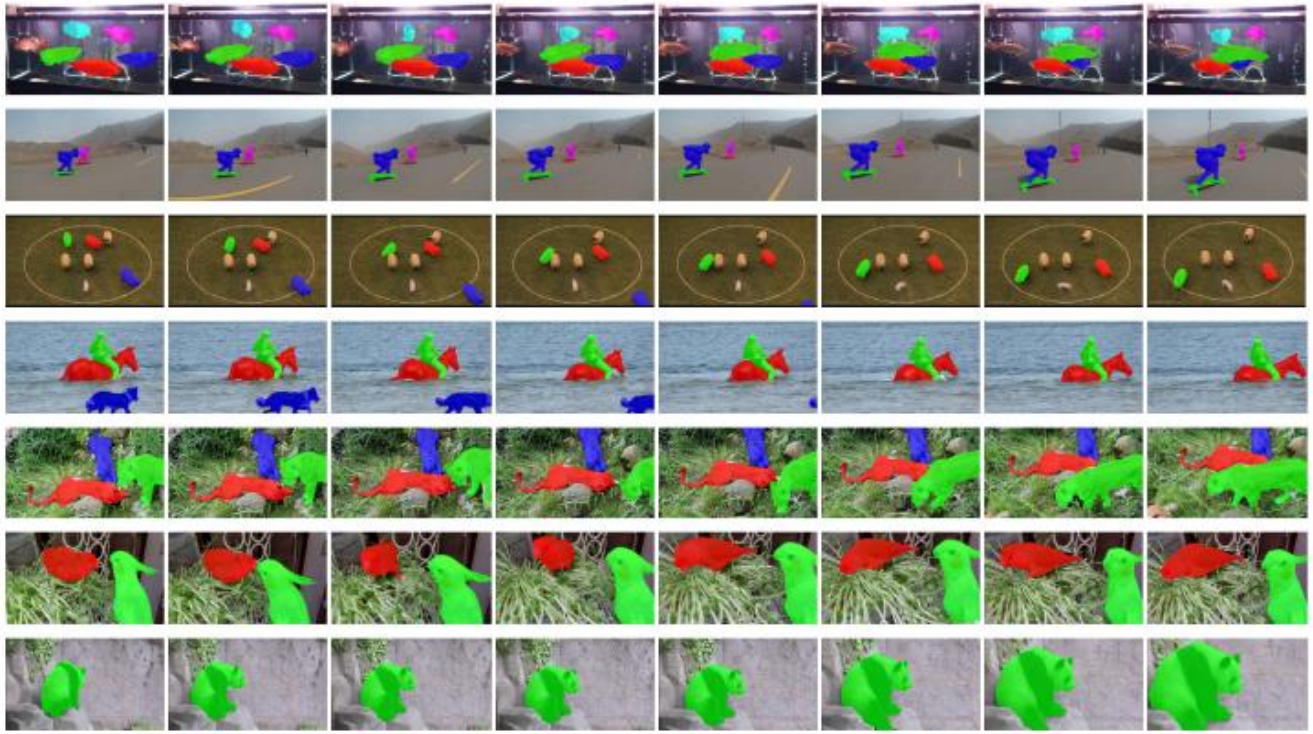
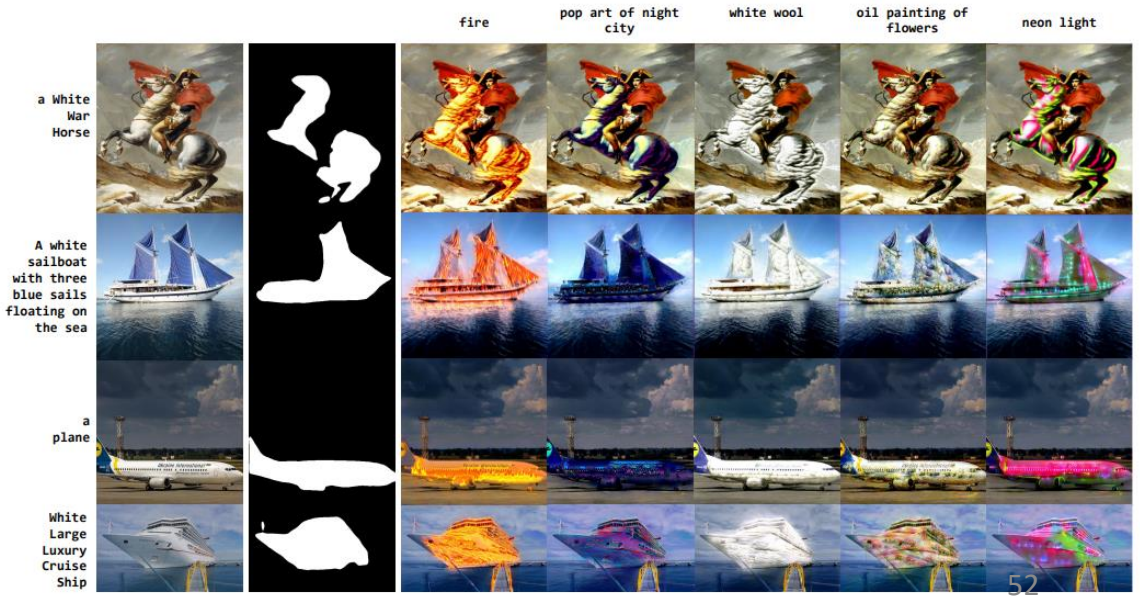


Figure 6. Qualitative results for one-shot video object segmentation on YouTube-VOS with multiple instances.



CLIPStyler training hours of other networks the parameters

CLIPStyler

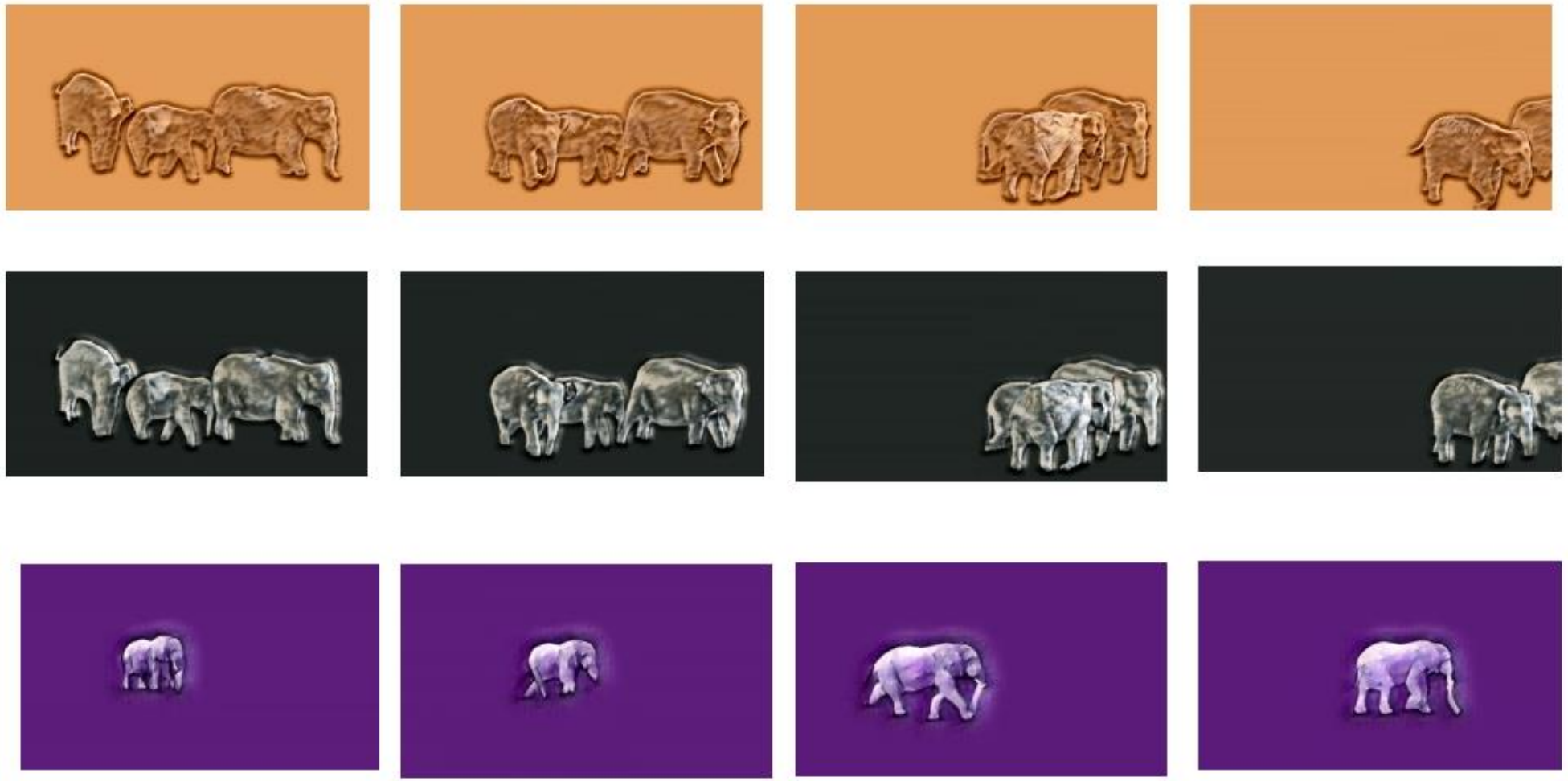


Figure 1. Our style transfer results on various text conditions. Translated images have spatial structure of the content images with realistic textures corresponding to the text.



Figure 5. Comparison with other text-guided manipulation models. Our results have more realistic textures on the entire location. Baseline models show limited change, and in some cases the contents of the image are modified.

CLIPStyler



CLIPStyler



Evaluation Approach I

Question	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
The stylized object matches the describe text condition	77.4%	19.4%			
The styled object shape remains consistent across every frame in the video	74.2%	25.8%			
The original shape and structure of the object preserved after applying the style	80.6%	19.4%			
The background is unaffected by the style applied	9.7%	35.5%	22.6%	22.6%	
The transferred style aligns with the object motion throughout the video	96.8%				

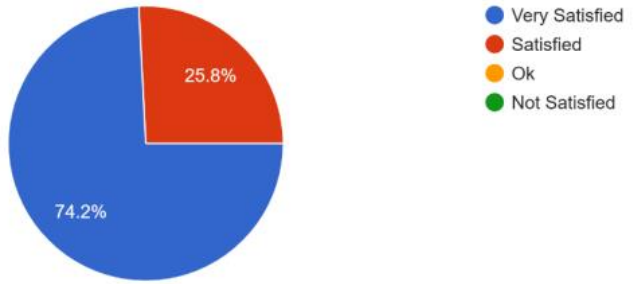


Figure 6.19:Chart showing the votes for the question "How satisfied are you with the styles transferred to multiple objects in the video"

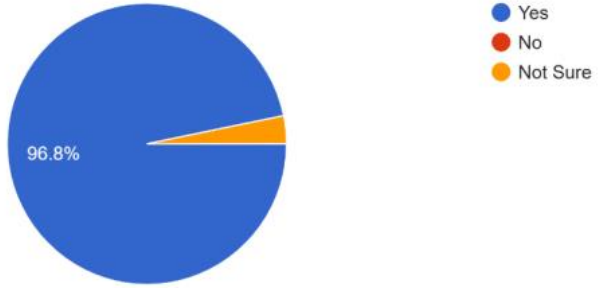
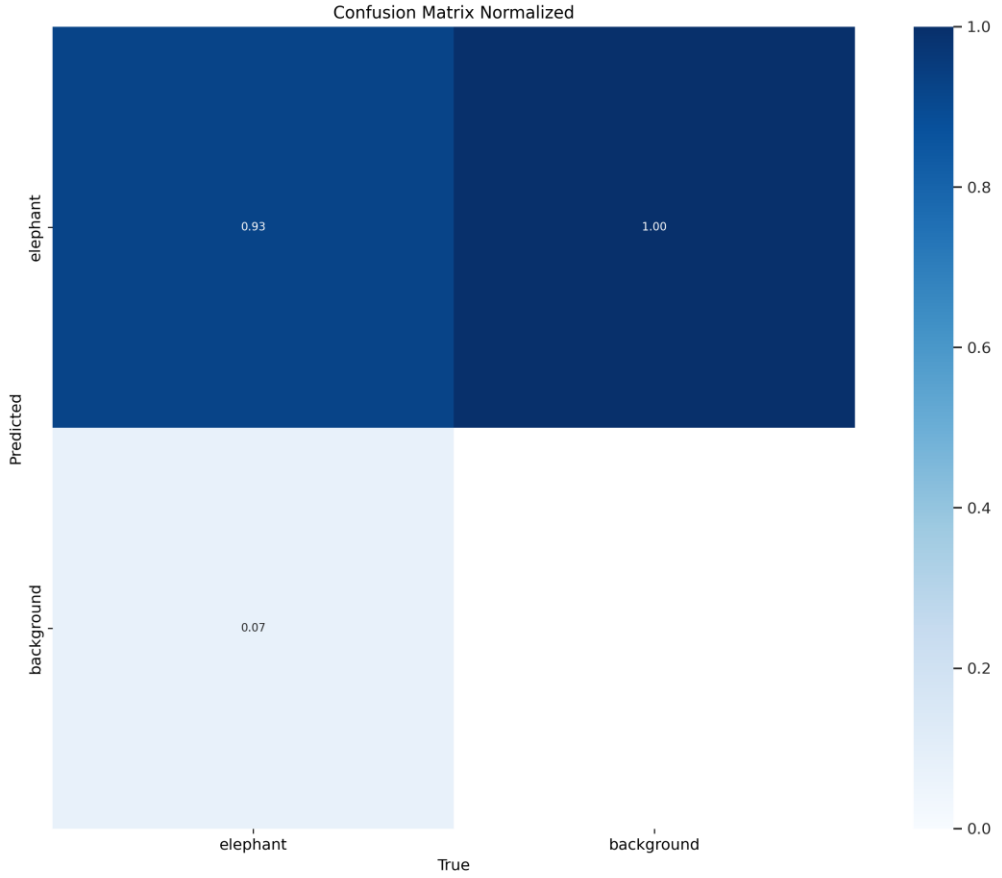
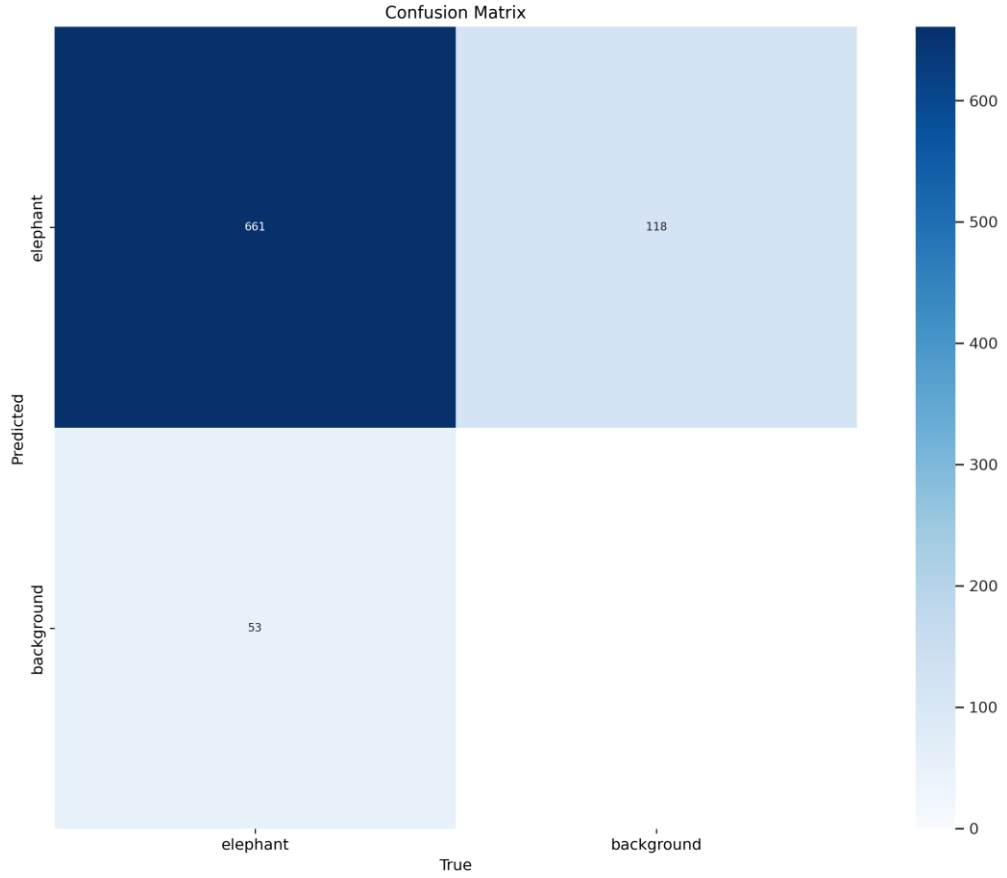


Figure 6.20:Chart showing the votes for the question "Does the transferred style align with the object's motion throughout the video"

YOLO TRAINING- best one validation





YOLO TRAINING- best one

YOLOv8m Training Configuration

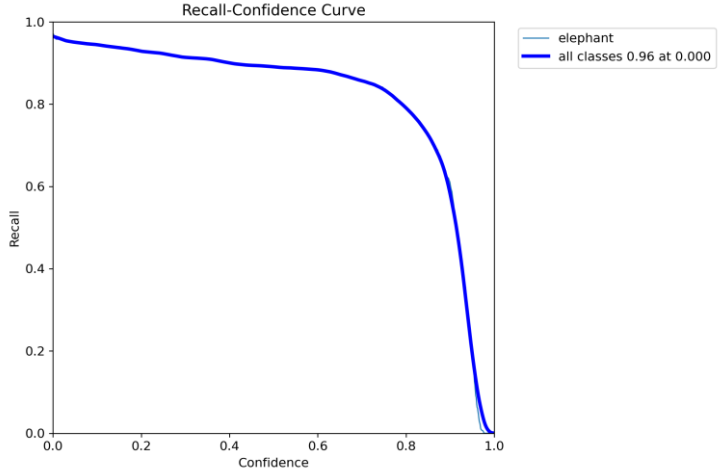
Model Setup
- Model: YOLOv8m (medium)
- Image Size: 800px
- Batch Size: 16
- Epochs: 400

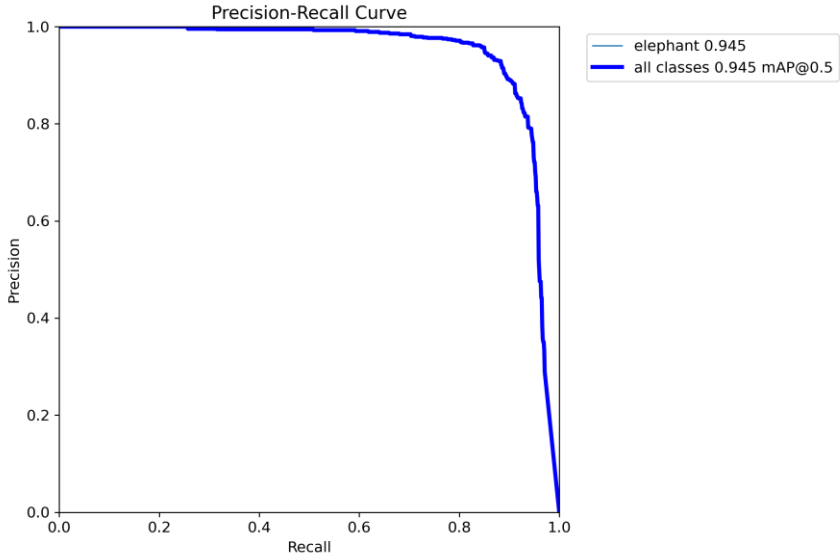
Optimization
- Initial Learning Rate: 0.0001
- Optimizer: Adam
- Weight Decay: 0.0005
- Warmup Epochs: 3.0

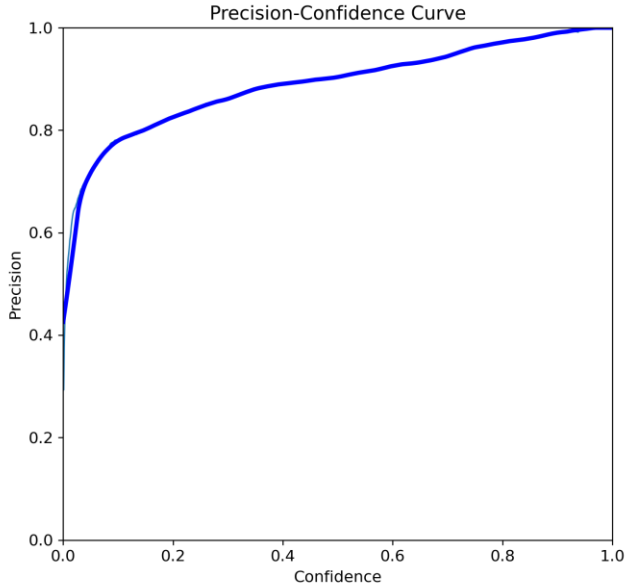
Loss Components
- Box Loss: 7.5
- Classification Loss: 0.5
- DFL Loss: 1.5

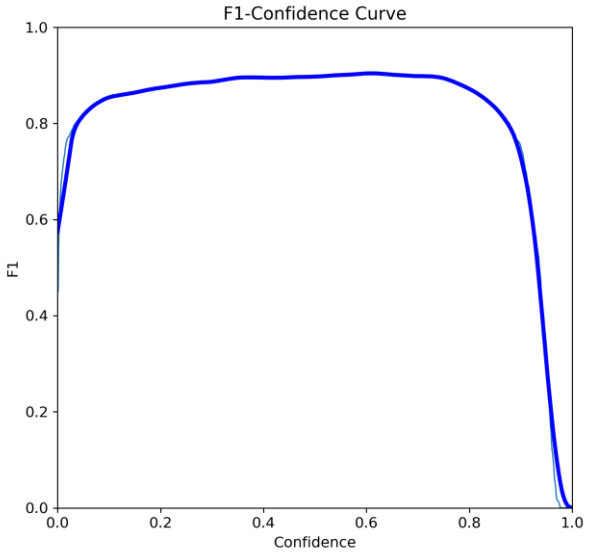
Detection Settings
- IOU Threshold: 0.7
- Confidence Threshold: 0.1
- Maximum Detections: 300

YOLO TRAINING- best one

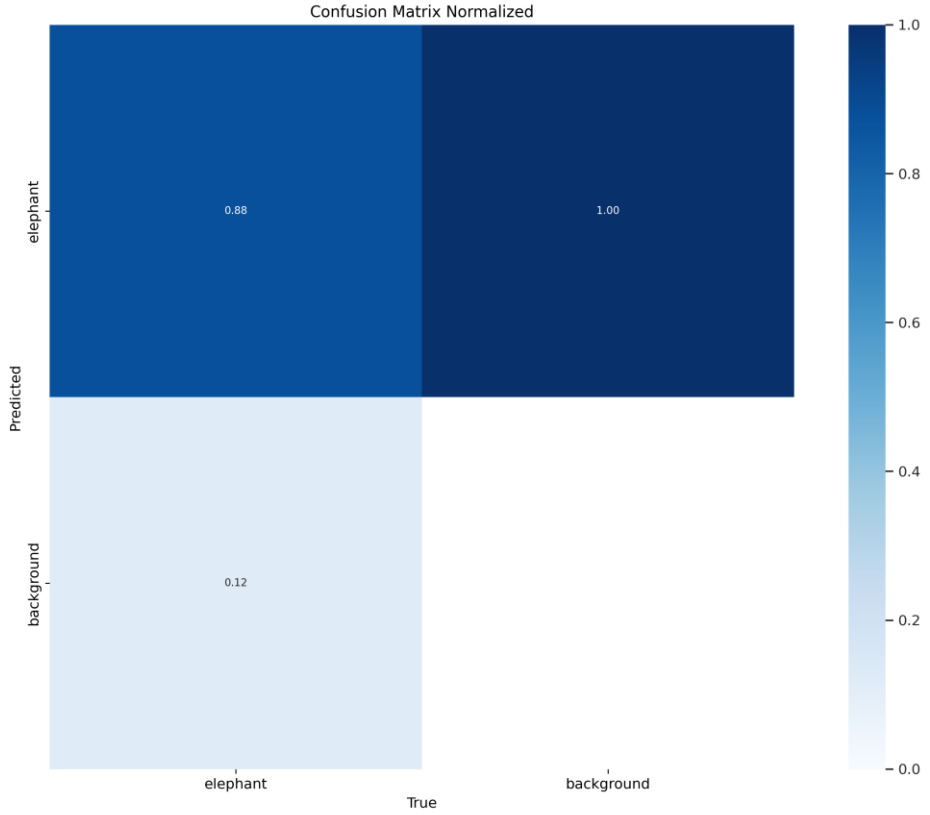


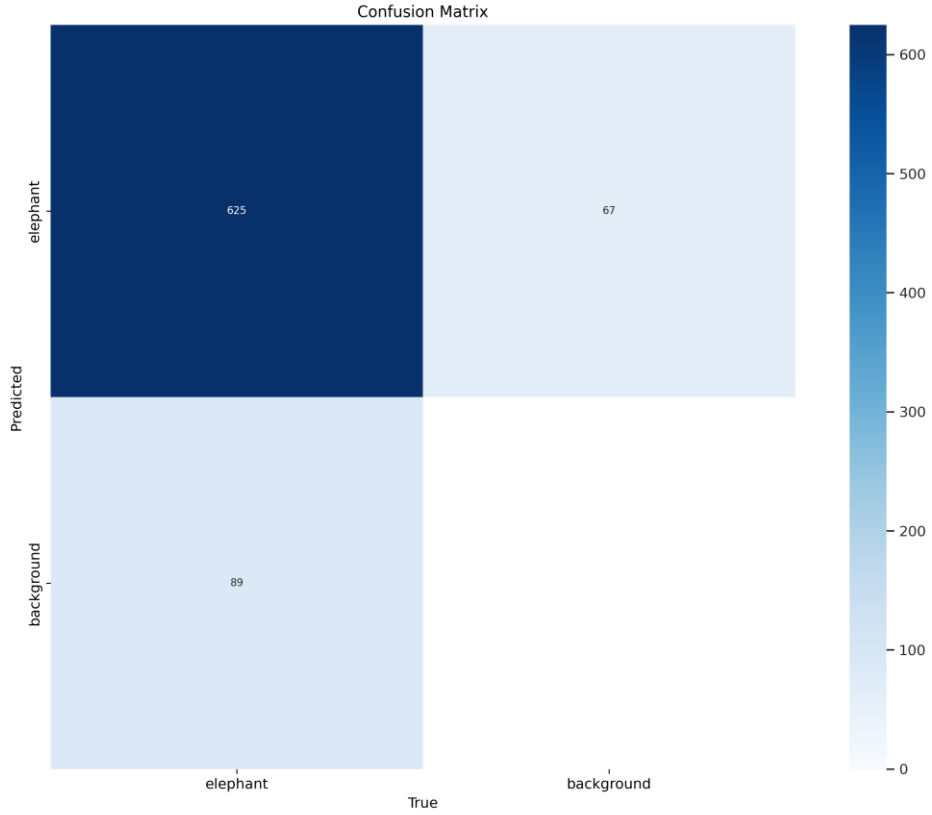






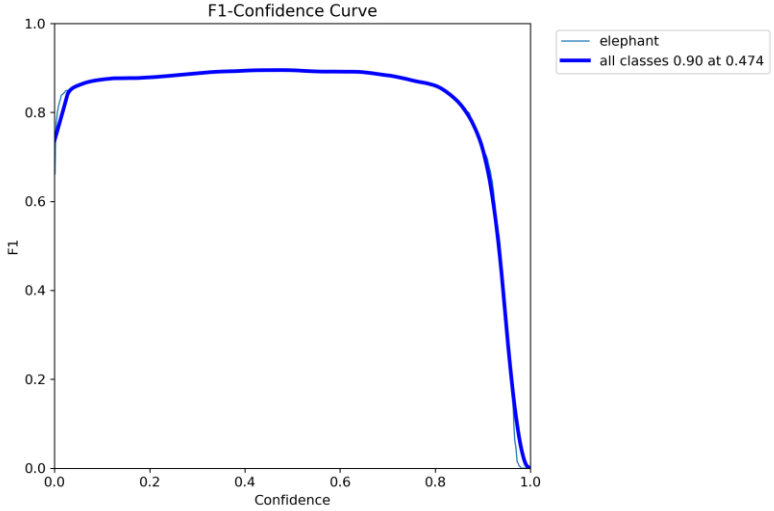
YOLO TRAINING- good one

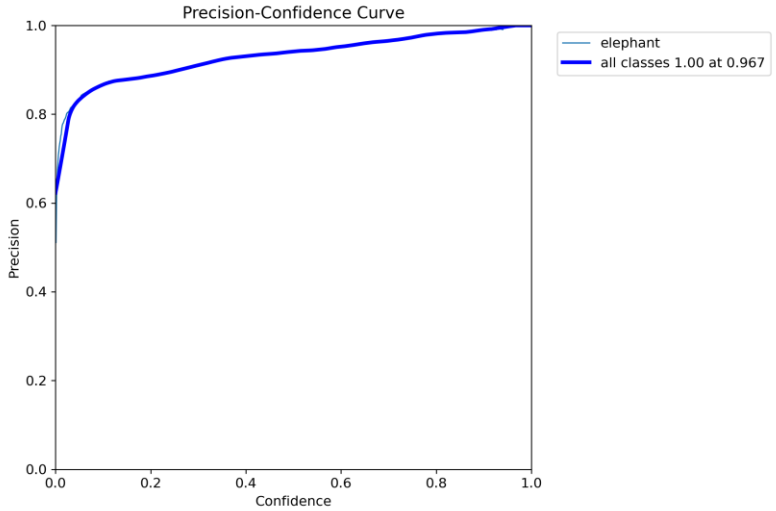


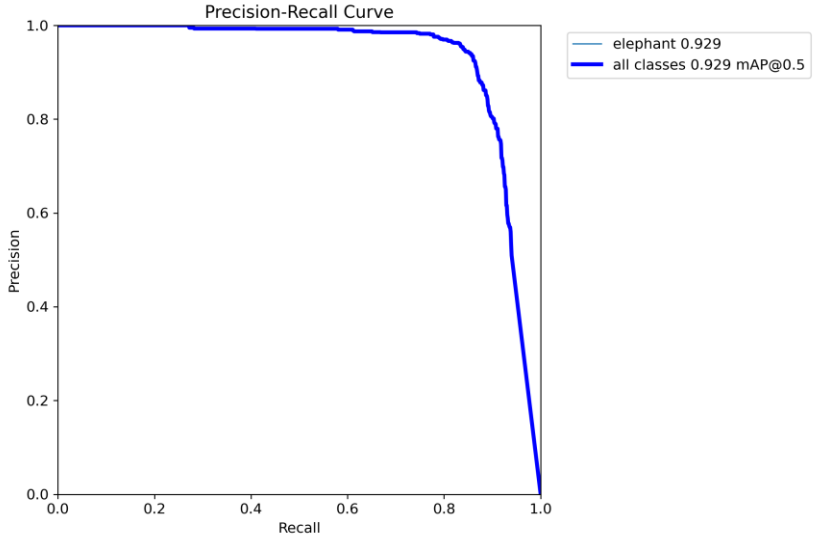


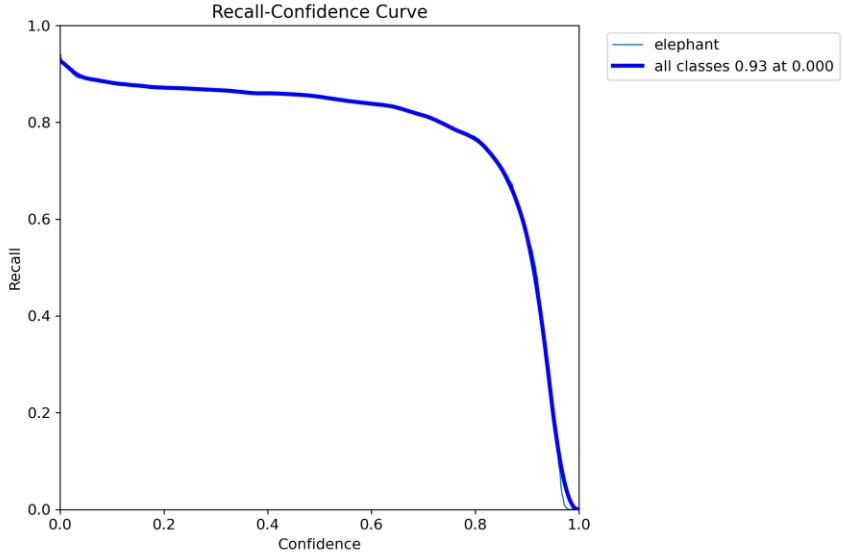
YOLO TRAINING- good one

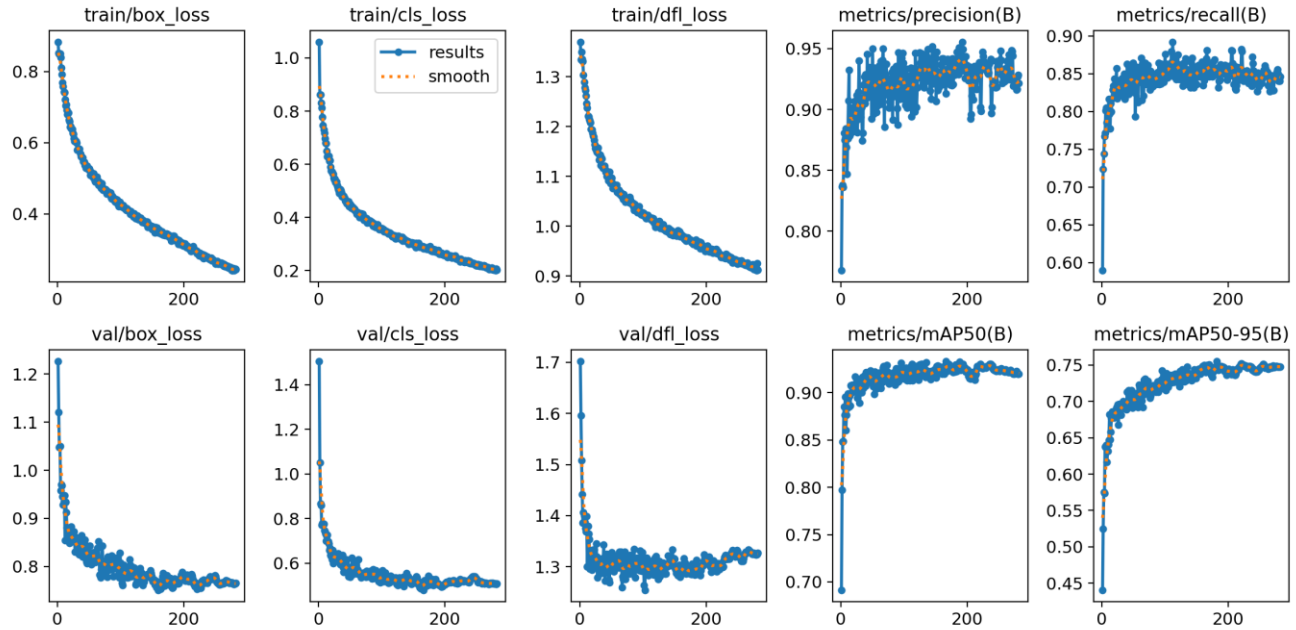
Model Setup
•Model: YOLOv8s
•Image Size: 800px
•Batch Size: 16
•Epochs: 300
Optimization
•Initial Learning Rate: 0.0005
•Optimizer: Adam
•Weight Decay: 0.0005
•Warmup Epochs: 3.0
Loss Components
•Box Loss: 7.5
•Classification Loss: 0.5
•DFL Loss: 1.5
Data Augmentation
•Mosaic: 1.0
•Random Flip (LR): 0.5
•Scale: 0.5
•Auto Augment: RandAugment


F1-Confidence Curve
elephant
all classes 0.90 at 0.474
F1
Confidence

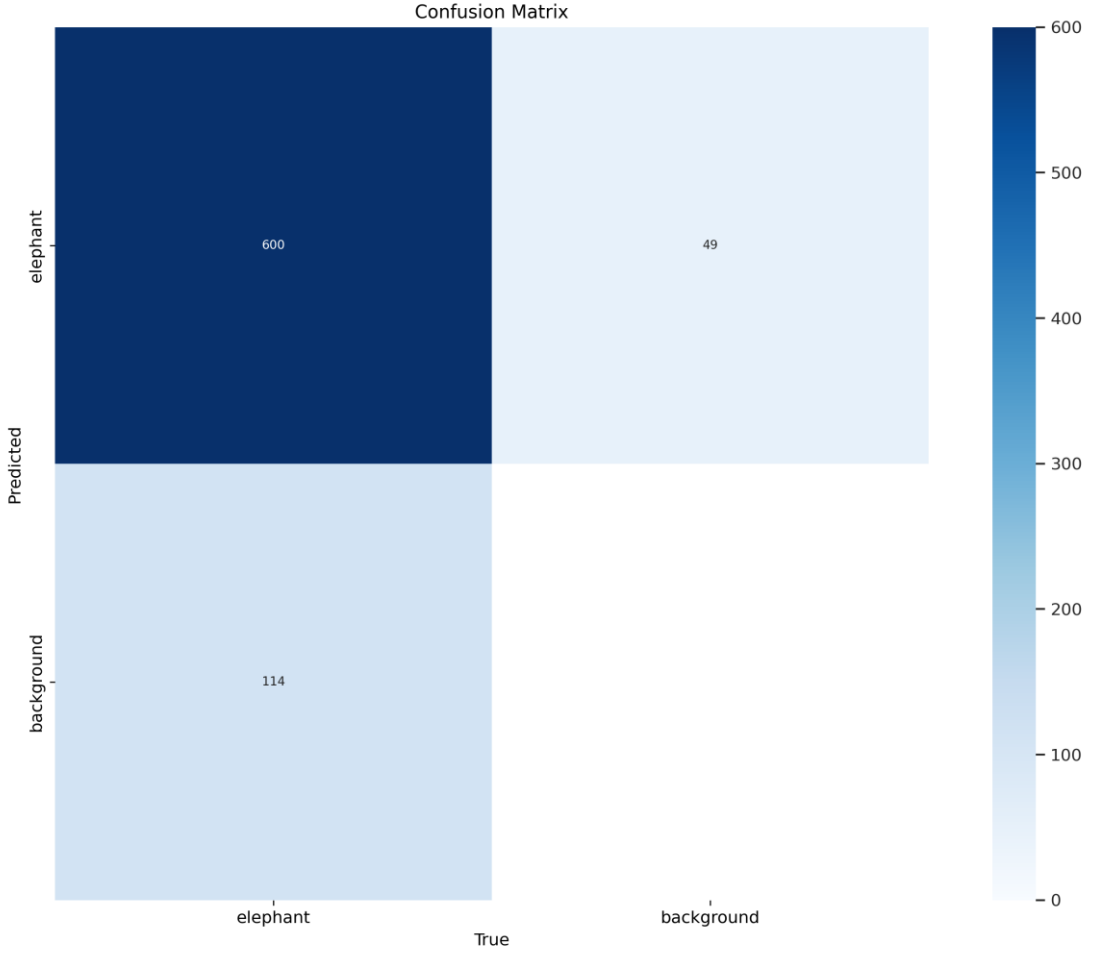

Precision-Confidence Curve
elephant
all classes 1.00 at 0.967
Precision
Confidence

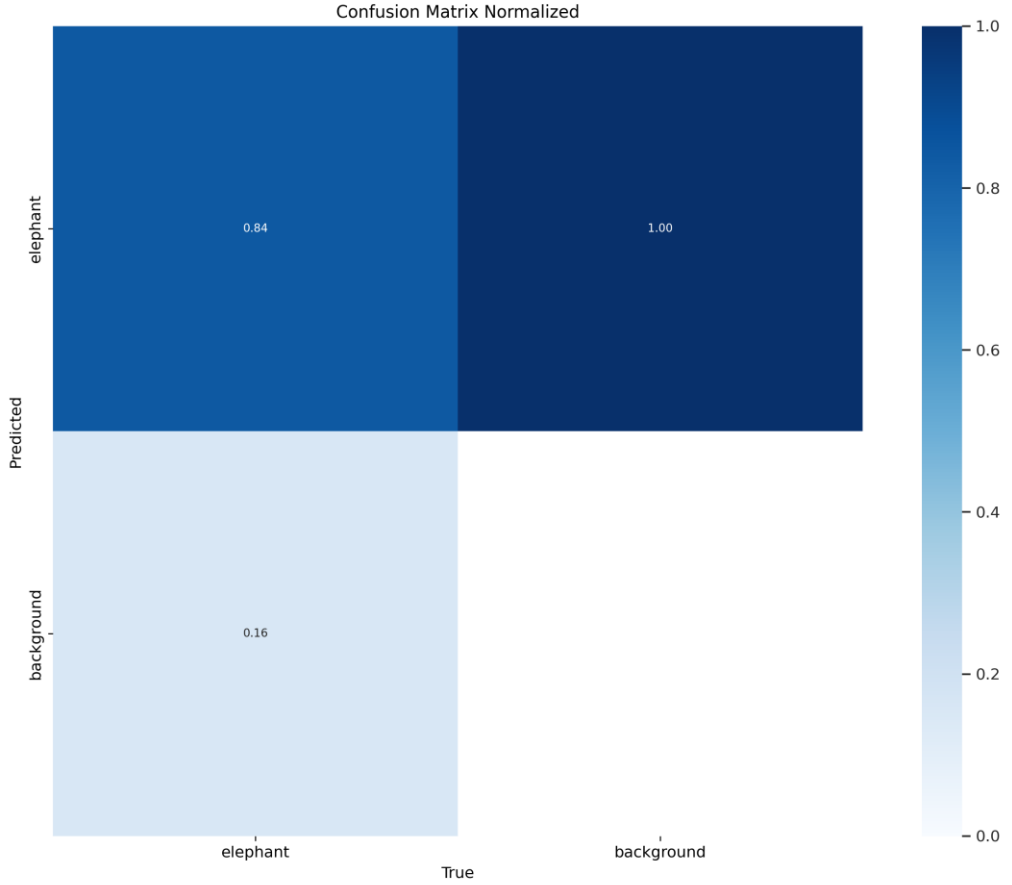

Precision-Recall Curve
elephant 0.929
all classes 0.929 mAP@0.5
Precision
Recall

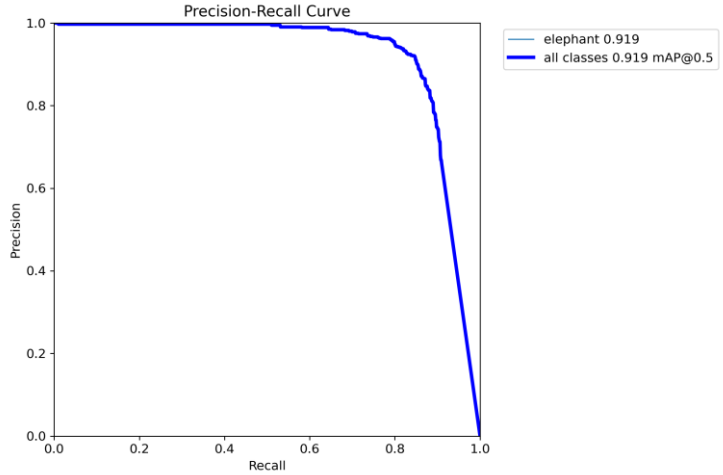

Recall-Confidence Curve
elephant
all classes 0.93 at 0.000
Recall
Confidence

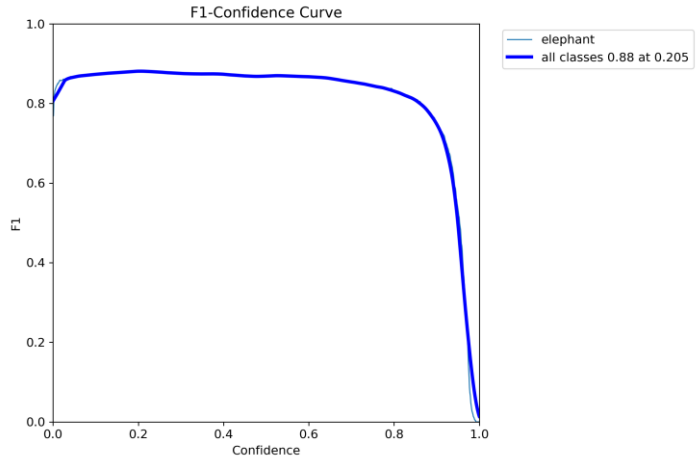

train/box_loss
train/cls_loss
results
smooth
train/dfl_loss
metrics/precision(B)
metrics/recall(B)
val/box_loss
val/cls_loss
val/dfl_loss
metrics/mAP50(B)
metrics/mAP50-95(B)

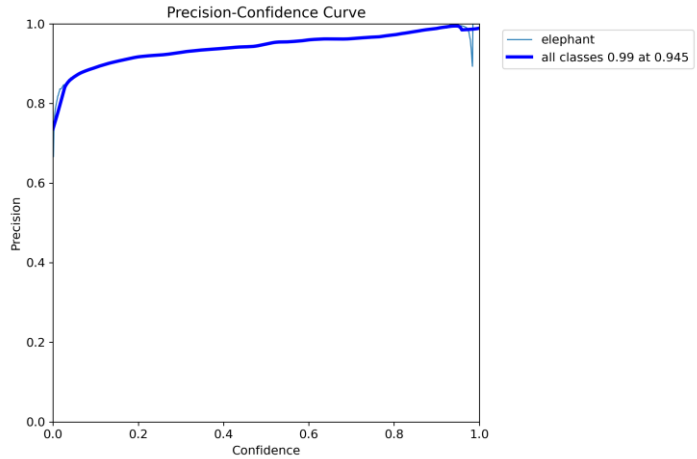
YOLO TRAINING- COLAB

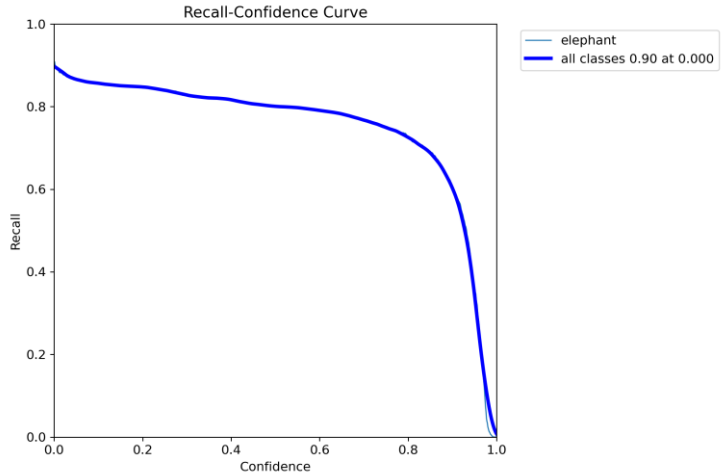


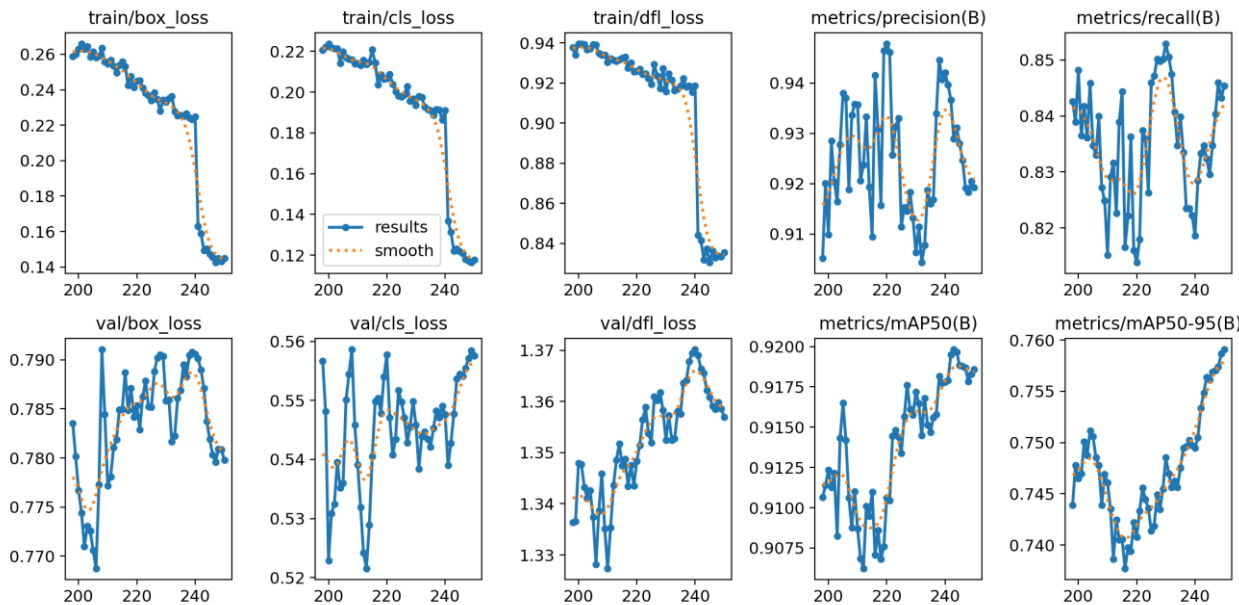



Precision-Recall Curve
elephant 0.919
all classes 0.919 mAP@0.5
Precision
Recall


F1-Confidence Curve
elephant
all classes 0.88 at 0.205
F1
Confidence


Precision-Confidence Curve
elephant
all classes 0.99 at 0.945
Precision
Confidence


Recall-Confidence Curve
elephant
all classes 0.90 at 0.000
Recall
Confidence


train/box_loss
train/cls_loss
train/dfl_loss
metrics/precision(B)
metrics/recall(B)
results
smooth
val/box_loss
val/cls_loss
val/dfl_loss
metrics/mAP50(B)
metrics/mAP50-95(B)

YOLO TRAINING- COLAB hyperparameters

Model: YOLOv8m
Image size: 800px
Batch size: 16
Epochs: 250

Initial Learning rate: 0.0005
Optimizer: Adam
Warmup Epochs: 3.0

Loss weights
Box loss: 7.5
Classification loss: 0.5
DFL loss: 1.5

Key Augmentations
Mosaic: 1.0
Scale: 0.5
Random Flip: 0.5

Parameter	Configuration 1	Configuration 2	Configuration 3
Model	YOLOv8m (medium)	YOLOv8s (small)	YOLOv8m (medium)
Epochs	250	300	400
Image Size	800px	800px	800px
Batch Size	16	16	16
Initial LR	0.0005	0.0005	0.0001
Optimizer	Adam	Adam	Adam
Box Loss Weight	7.5	7.5	7.5
Class Loss Weight	0.5	0.5	0.5
DFL Loss Weight	1.5	1.5	1.5
Warmup Epochs	3.0	3.0	3.0
Weight Decay	0.0005	0.0005	0.0005